

Discrete Response, Time Series and Panel Data - Notes

Nils Rechberger

2026-02-26

W-01-Timeseries-Introduction-Visualisation-Stationarity

 Note

Coming soon...

W-02-Timeseries-Introduction-Visualisation-Stationarity

Log-transformations

Instead of the original time series X_t , consider the log-transformed time series with the elements.

$$g_t = \log(X_t)$$

When to apply it?:

- When the distribution of the time series elements is right-skewed.
- When a time series could be produced by the product of model outcomes.

Differencing

Let's take a linear function (defined on a set of integer numbers)

$$X_t = m \cdot t + b$$

Then, the backward difference

$$X_t - X_{t-1} = (m \cdot t + b) - (m \cdot (t-1) + b) = m$$

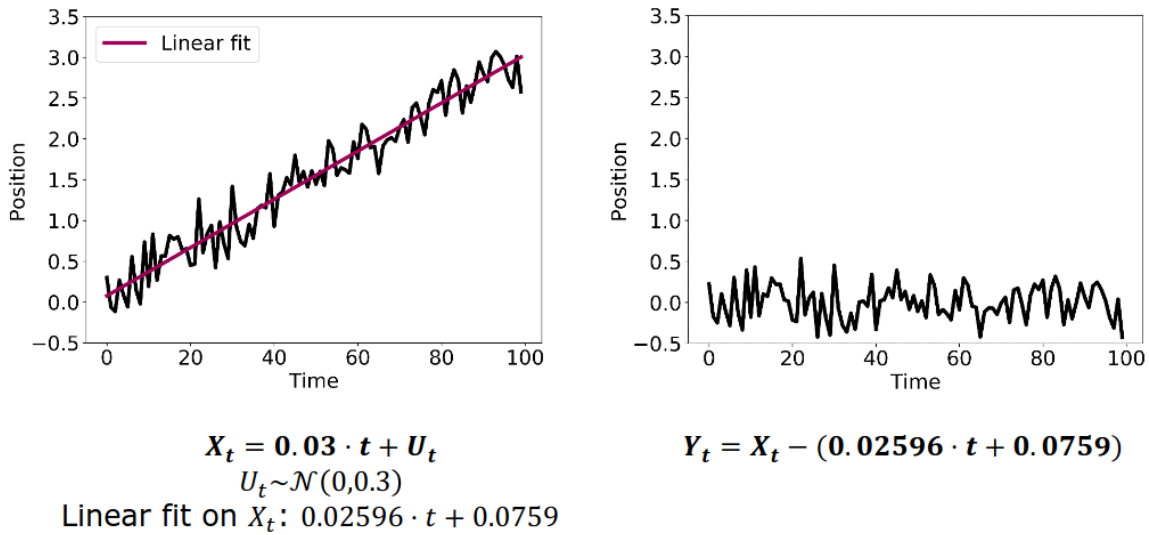
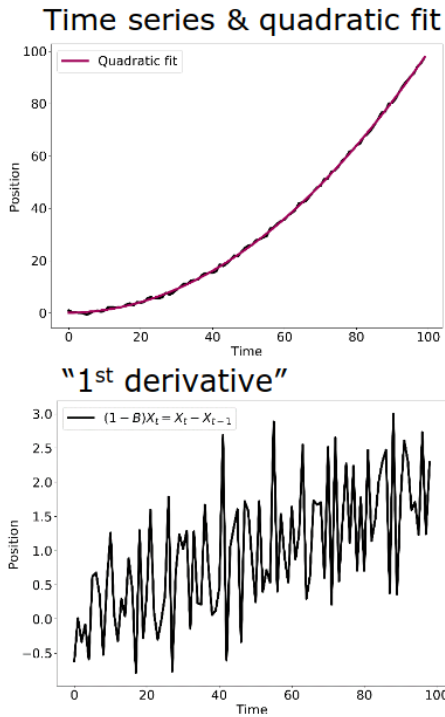


Figure 1: Differencing

Higher Order Differencing

Use multiple derivatives to eliminate lower-order terms. Trend order is reduced, but the stationary part may be more complex. After trend is eliminated, maybe a new dependencies in the stationary part.

Example



Initial time series: $X_t = 0.01 t^2 + U_t$ with $U_t \sim \mathcal{U}(-1,1)$

"1st derivative" displays still linear dependence, where as "2nd derivative" is "constant."

Note that a residual process does not have structure of U_t .

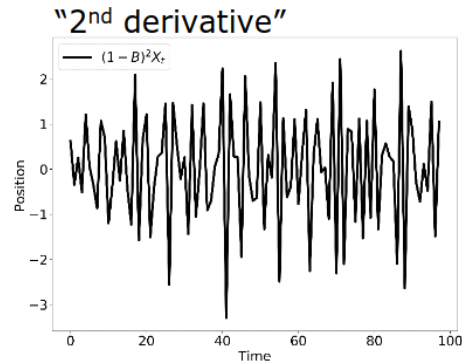
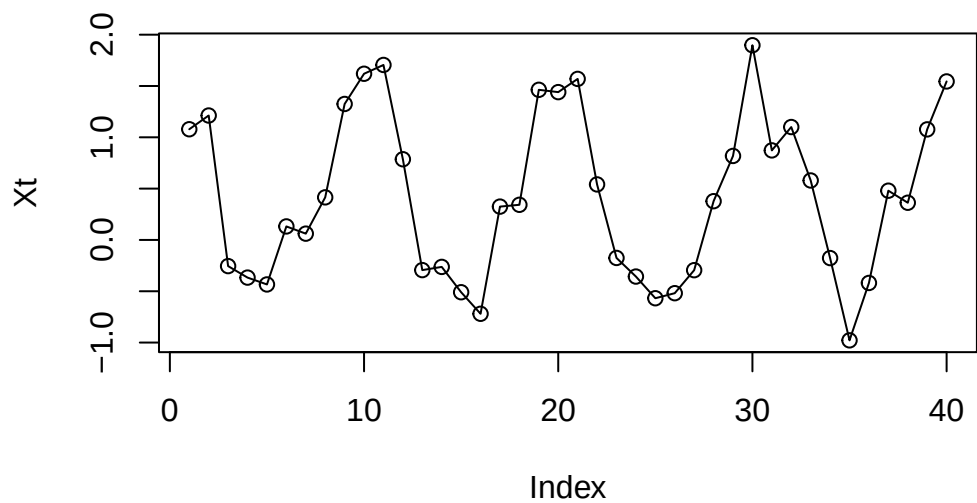


Figure 2: Remove Higher Order Trend

Differencing Seasonality

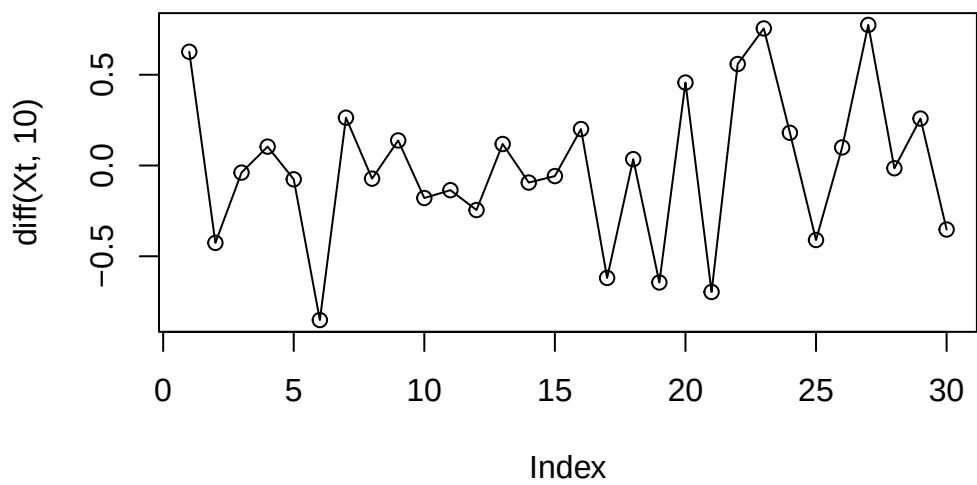
Seasonality manifests itself by a repetitive behavior after a fixed lag time p .

```
tRange<-1:40
Xt<-cos(2*pi/10*tRange)+runif(40)
plot(Xt,type="o")
```



What happens if we differentiate now with lag 10?

```
plot(diff(Xt,10),type="o")
```



Summary of differencing

Advantages:

- can remove trends and seasonality effects
- is easy to use

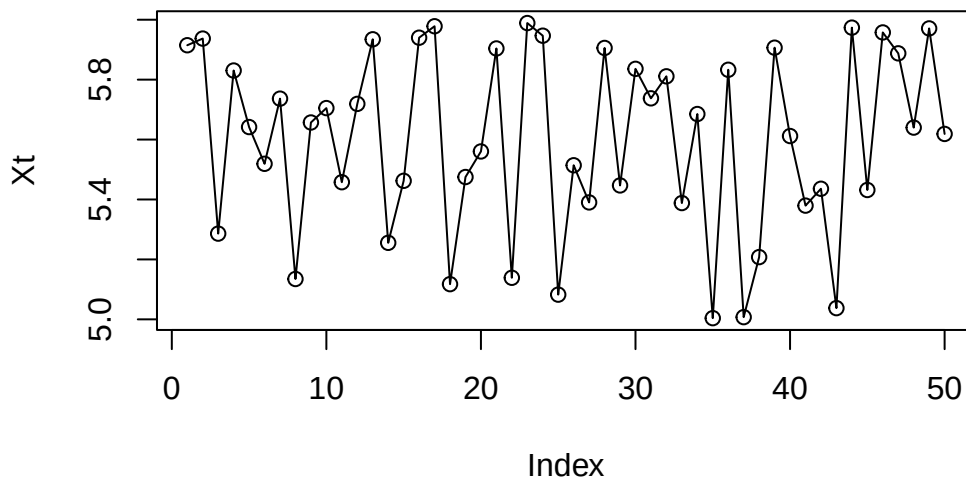
Disadvantages:

- the resulting time series is shorter than the original one
- the results of differencing may display dependencies among different elements
- the original decomposition elements are unknown

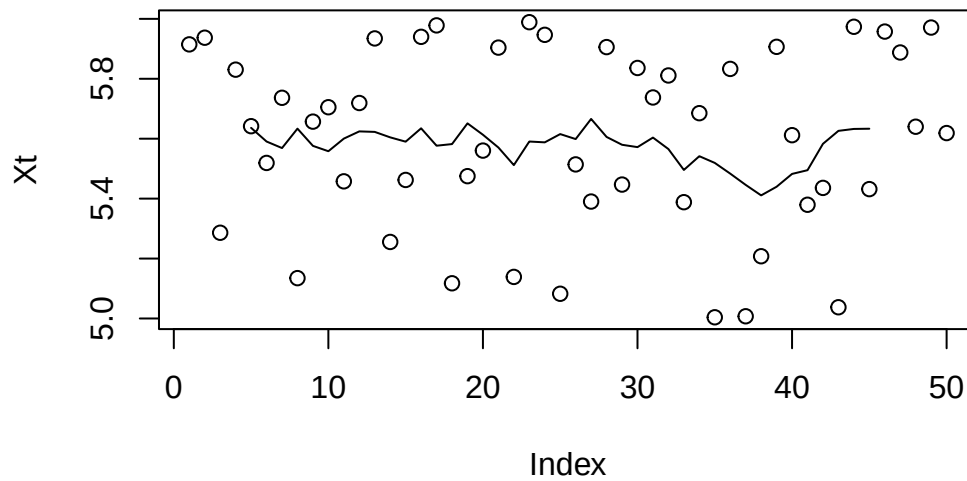
Filtration

To reduce the noise element in an experiment, you can repeat the measurement multiple times and then calculate the average.

```
set.seed(42)
Xt<-5+runif(50)
plot(Xt,type="o")
```



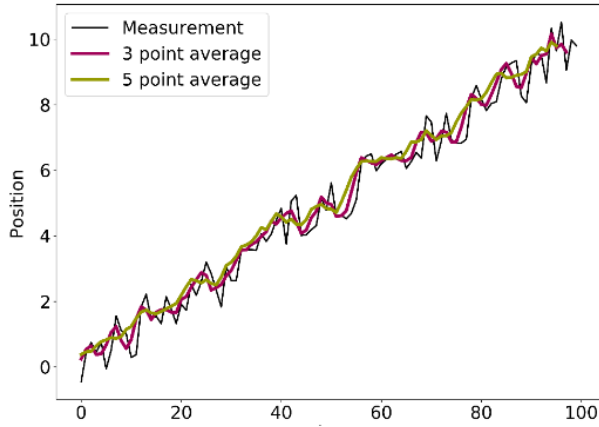
```
filtered<-filter(Xt,filter=rep(1,10)/10)
plot(Xt)
points(filtered,type="l")
```



Reasons for using Filtration

By averaging measurements, we obtain a new averaged time series. By increasing the number of averaged measurements, the parts converges to the expectation value of the measurement error, i.e. 0.

Note: The short-term variations are smoothed out and the trend is confirmed.



Initial time series:

$$X_t = 0.1 t + U_t, U_t \sim \mathcal{U}(-1,1)$$

$$\text{3 point average: } a_0 = a_{-1} = a_{-2} = \frac{1}{3}$$

$$\text{5 point average: } a_0 = a_{-1} = a_{-2} = \frac{1}{5} \\ = a_{-3} = a_{-4} = \frac{1}{5}$$

Figure 3: Filtering

Linear filters to quantify seasonality effects

In case of seasonality effects, the filter has to expand over the full period to extract the trend. To quantify the seasonal impact, the values of the same time in the season have to be averaged.

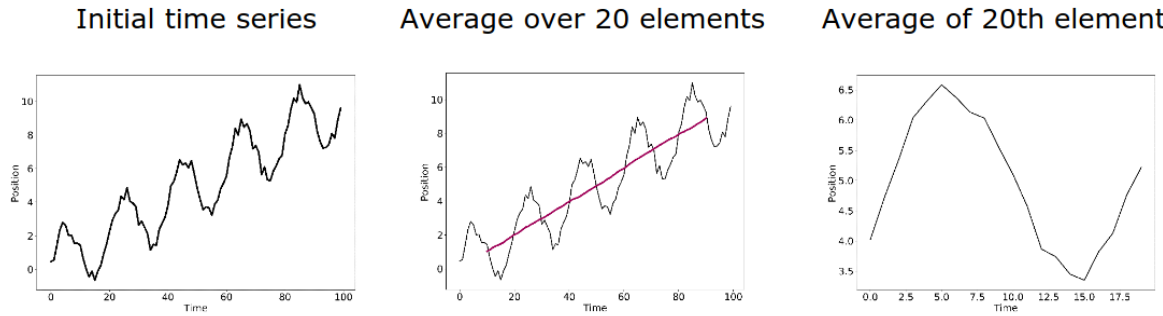


Figure 4: Quantify Seasonality Effects

Seasonal-trend decomposition by Loess Algorithm (STL)

Linear filters are great if periodicity is obvious; but in less obvious cases, extracting the individual components may be laborious. An alternative is the seasonal-trend decomposition procedure by Loess: An iterative, non-parametric smoothing algorithm that is more robust.

Fitting of Periodic Function

If you know the generative model, you can use fitting to determine the contributions of trend and seasonality. Fitting methods (among others):

- Ordinary least square (OLS)
- Robust fitting (to prevent the impact of outliers)
- GLS (generalised least square) / time series regression methods

```
set.seed(42)
#Generate the sample data set
t<-seq(1,100,length=100)
data<-0.1*t+cos(2*pi/10*t)+runif(100)
ts<-ts(data)
#Fit the model
fit<-lm(ts~t+cos(2*pi/10*t))
summary(fit)
```

Call:

```
lm(formula = ts ~ t + cos(2 * pi/10 * t))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.58266	-0.22271	0.04478	0.25411	0.49285

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.639744	0.059840	10.69	<2e-16 ***
t	0.097718	0.001029	94.98	<2e-16 ***
cos(2 * pi/10 * t)	0.973247	0.041999	23.17	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2969 on 97 degrees of freedom

Multiple R-squared: 0.9901, Adjusted R-squared: 0.9899

F-statistic: 4836 on 2 and 97 DF, p-value: < 2.2e-16

Autocorrelation

The autocorrelation depends only on the lag for weakly or strongly stationary time series.

$$p(k) := \text{Cor}(X_{t+k}, X_t)$$

Interpretation of autocorrelation

The correlation $p(k)^2$ squared corresponds to the percentage of variability explained by the linear association between X_t and X_{t+k} .

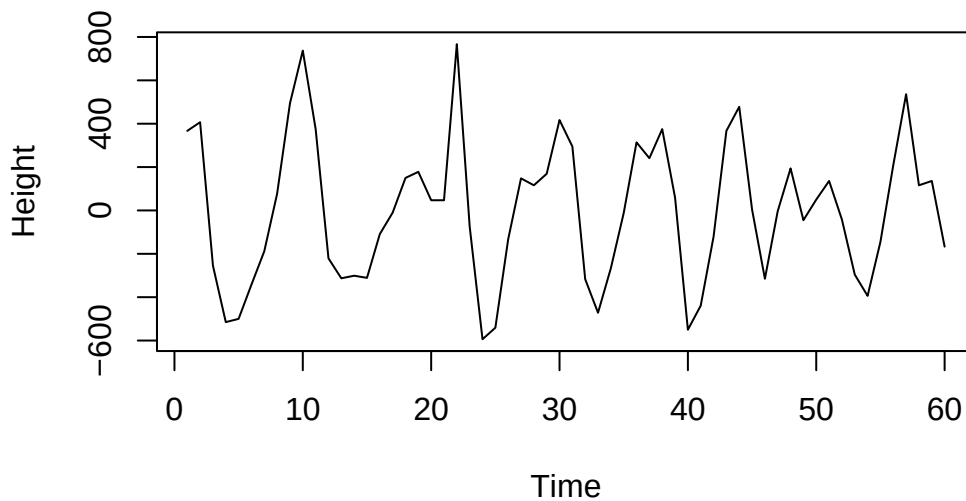
Lagged Scatter Plot

Exchange average over realizations with average over time.

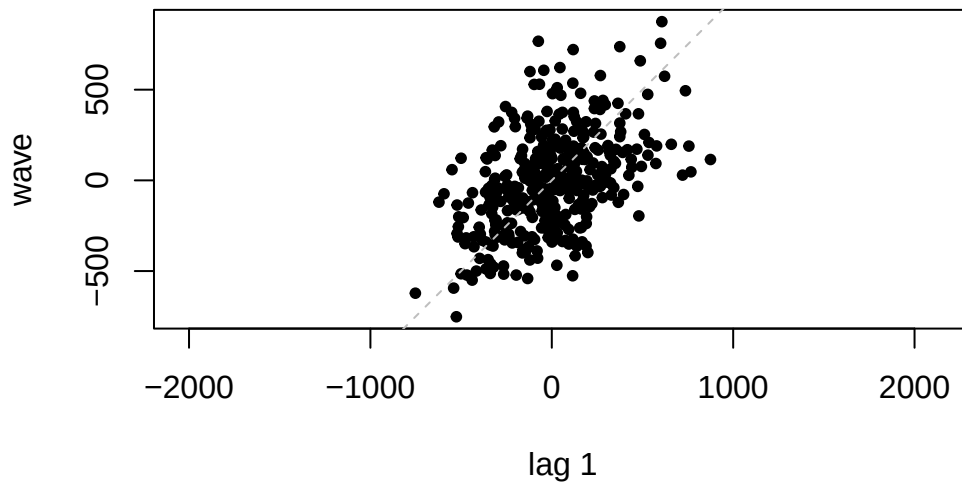
```
address <- "https://raw.githubusercontent.com/AtefOuni/ts/master/Data/wave.dat"
dat <- read.table(address,header=T)

#Generate ts object
wave<-ts(dat$waveht)

#Visualize the data
plot(window(wave,1,60),ylab="Height")
```

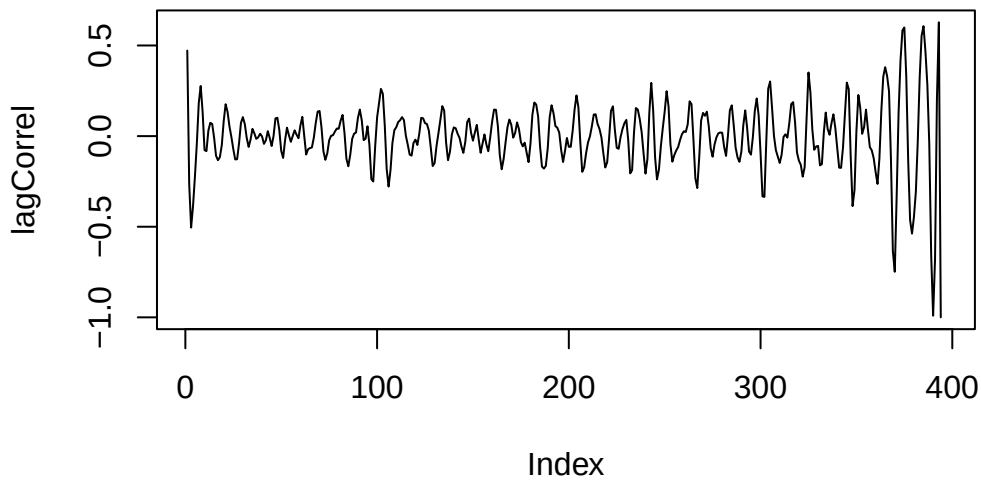


```
#Lag plot  
lag.plot(wave,do.lines=FALSE,pch=20)
```



Calculate lagged scatter plot estimator

```
# Calculate the Pearson Correlation coefficients  
n <- length(wave)  
lagCorrel <- rep(0,n)  
  
for (i in 1:(n-1)){  
  lagCorrel[i]=cor(wave[1:(n-i)],wave[(i+1):n])  
}  
  
plot(lagCorrel,type="l")
```



Plug-In Estimation

The plug-in estimator is the standard approach to computation autocorrelations. Calculated with an `acf` function in R.

Summary Autocorrelation

- (Auto)correlation measures the linear association at a given lag
- For realizations, the autocorrelation is estimated by running the calculations over time instances instead of realizations (therefore stationarity is desired)
 - Lagged scatter plot estimator
 - Plug-in estimator
- For high lags, the lagged scatter plot estimator diverges (because too few time points are considered)
- The plug-in estimator corrects the diverging behavior but generates a bias.

⚠ Warning

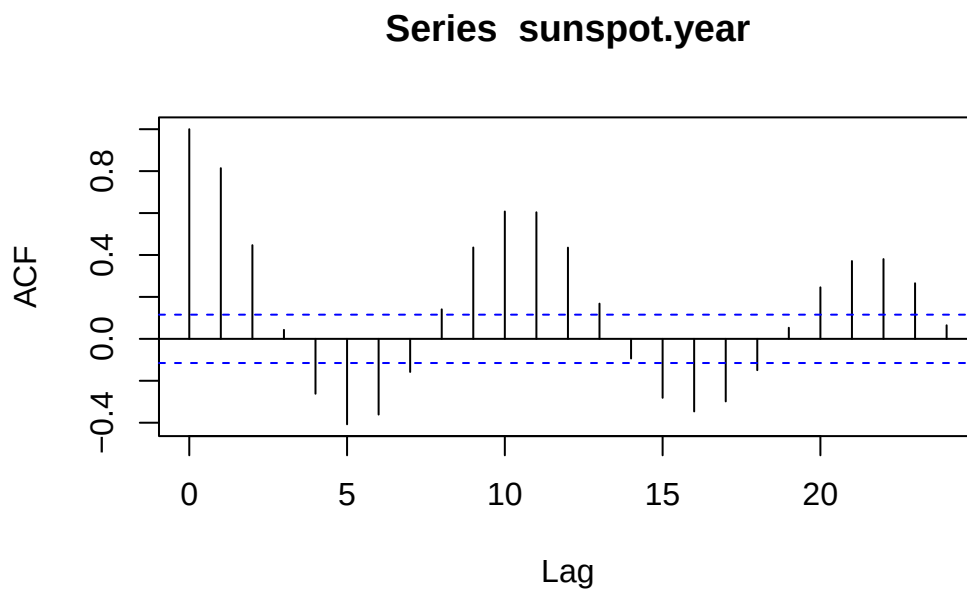
Warning: Non-linear relations may confuse the autocorrelation.

W-03-Timeseries-ACF-AR

Correlogram

Standard command: `acf(data)` for plug-in estimator values.

```
data(sunspot.year)
acf(sunspot.year)
```



Caution

The first lag at lag 0 always have to be a value of 1

Bounds of Autocorrelation

Under the null hypothesis of a time series process with iid distributed random variables, a 95 % acceptance region for the null hypothesis is:

$$\pm \frac{1.96}{\sqrt{n}}$$

For stationary processes, plug-in estimator value $\hat{p}(k)$ within the confidence band $\pm \frac{1.96}{\sqrt{n}}$ are considered to be different from 0 only by chance.

Ljung-Box Test

The Ljung-Box test assesses the null hypothesis that the random variables are independent when the h autocorrelation coefficients $\hat{p}(k)$ for $k = 1, \dots, h$ are equal to zero.

```
Box.test(sunspot.year, lag=10, type="Ljung-Box")
```

Box-Ljung test

```
data: sunspot.year  
X-squared = 542.41, df = 10, p-value < 2.2e-16
```

Characteristic Patterns in ACF

Patterns in ACF for non-stationary time series

For non-stationary time series, the ACF decays only slowly.

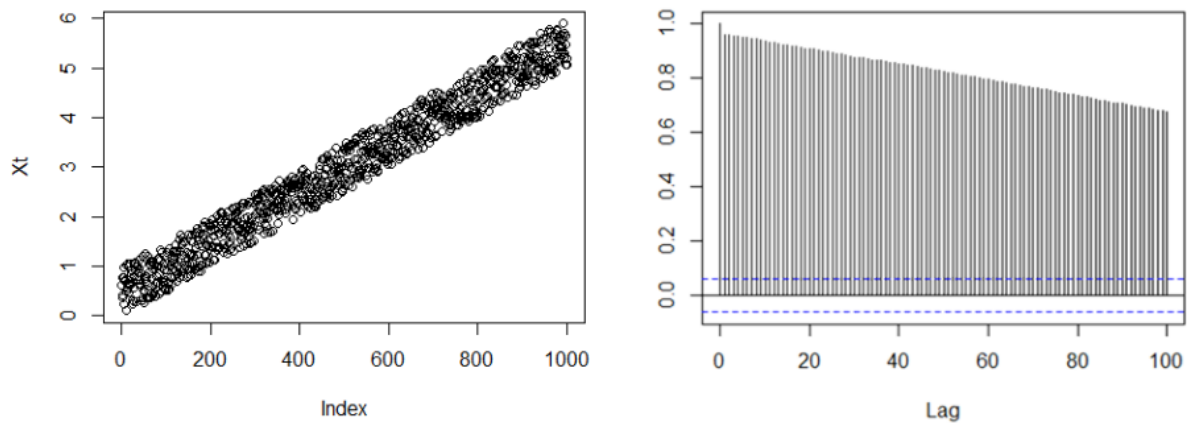


Figure 5: Non-Stationary time series

ACF of time series with seasonal pattern

For time series with seasonality, the ACF displays periodic structure.

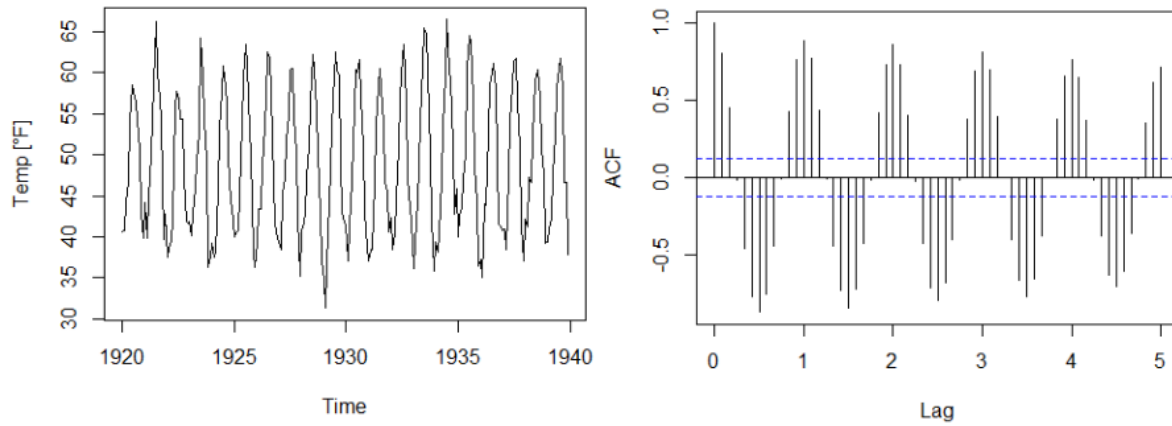


Figure 6: Time series with seasonal pattern

ACF of time series with seasonal pattern and trend

Time series with seasonality and trend, the ACF displays periodicity and slow decay.

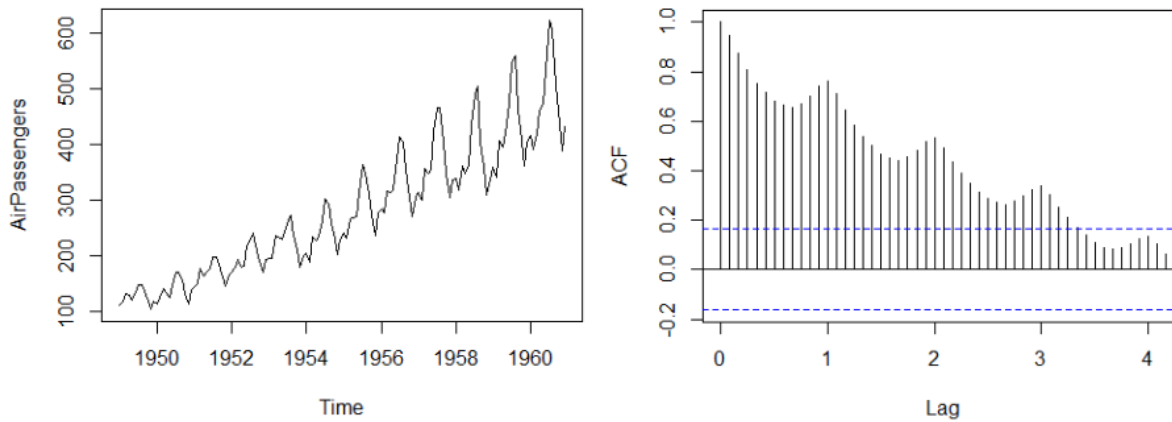


Figure 7: Time series with seasonal pattern and trend

ACF of time series with outliers

Outliers may be diagnosed by lagged scatter plots. Note that each outlier causes other outliers.

! Important

The plug-in estimator $\hat{p}(k)$ is sensitive to outliers.

Properties of ACF

Autocorrelation of process vs. estimated values of realization

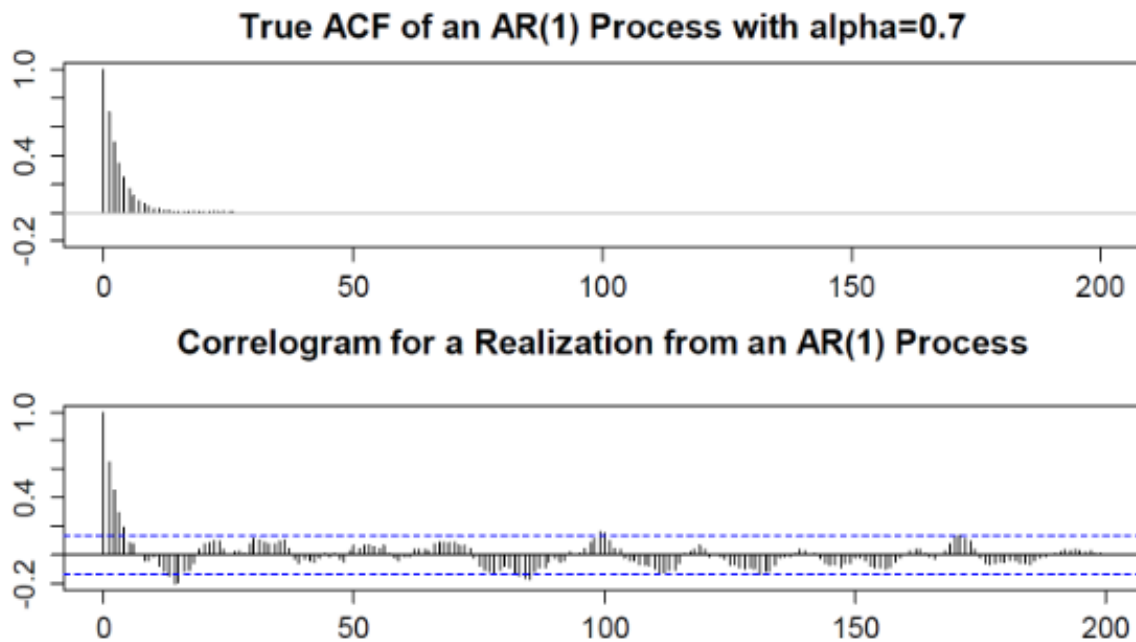
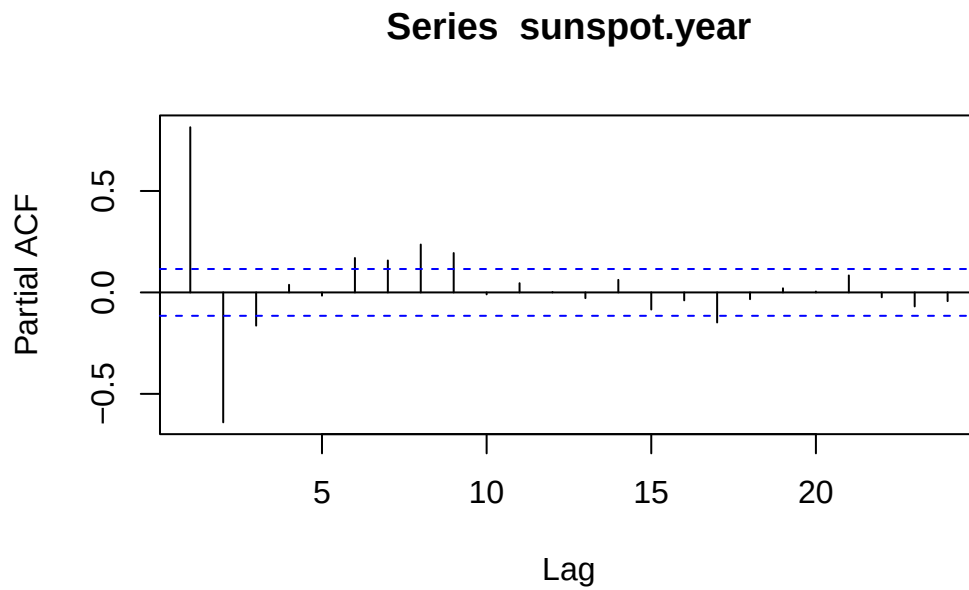


Figure 8: Correlogram of one realization compare with the true plot

Partial Autocorrelation

A challenge in the interpretation of ACFs is that the effect of smaller lags remains. The partial autocorrelation function (PACF) gives the partial correlation of a stationary time series with its own lagged values, regressed the values of the time series at all shorter lags.

```
pacf(sunspot.year)
```



W-04-Timeseries-ACF-AR

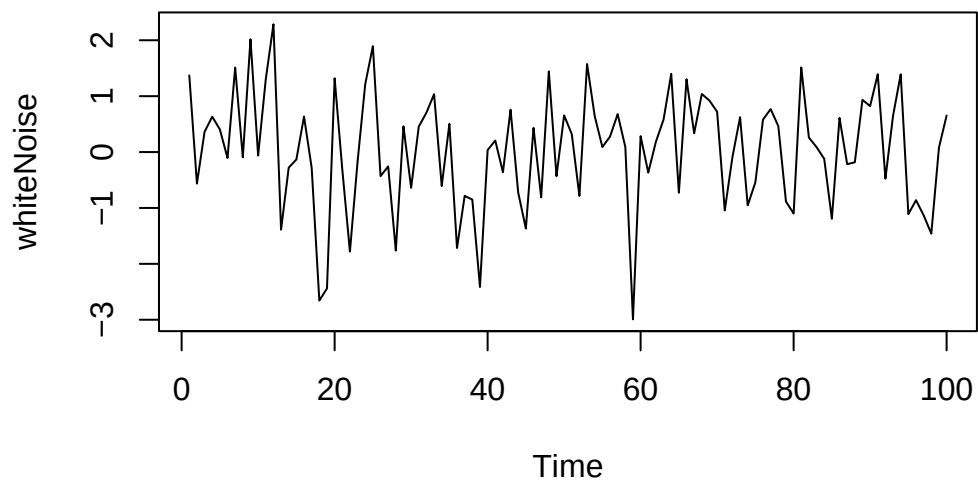
Fundamentals of Modeling

White noise

A time series process $W_t, t \in \mathbb{N}$ is called white noise if it is a sequence of independent and identically distributed random variables with mean 0.

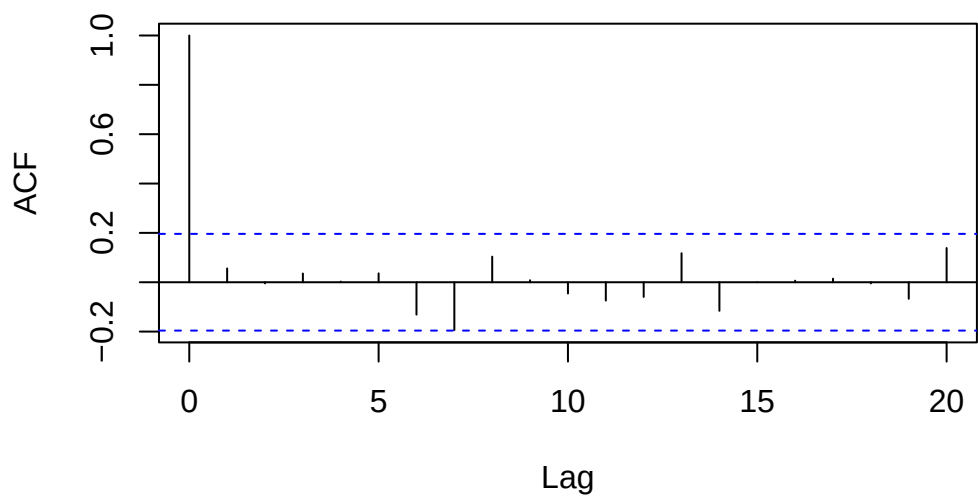
Note: The name “white” indicates that all frequencies are included, as in the case of “white light.”

```
set.seed(42)
whiteNoise <- ts(rnorm(100,mean=0,sd=1))
plot(whiteNoise)
```

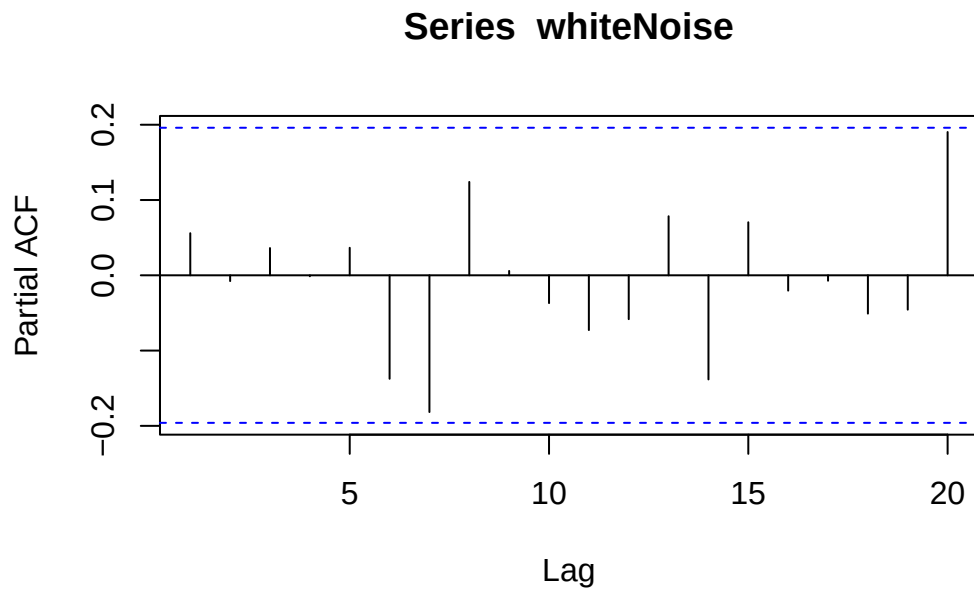



```
acf(whiteNoise)
```

Series whiteNoise



```
pacf(whiteNoise)
```



Causal White Noise

A series of *iid* random variables e_t , which are individually independent of the proceeding time series elements X_t

Basic Time Series Models

The following models are often used when there is limited information about how time series data is generated:

- AR: auto-regressive model
- MA: moving average model
- ARMA: combination of AR and MA model
- ARIMA: non-stationary ARMA model (I=integration)
- SARIMA: seasonal ARIMA model

Auto-Regression Models (AR)

Express the current random variable as a linear combination of the last p time steps and a “completely independent” term e_t called innovation.

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t$$

where α_x is a set of finite parameters and e_t a series of independent random variables.

$AR(1)$ as Random Walk Model

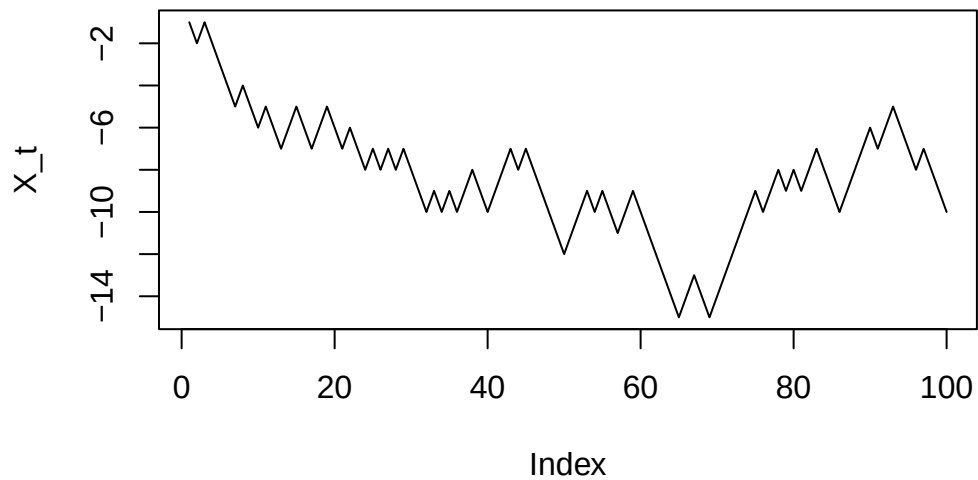
Trajectory of an object moving along a line. At each moment, the object takes randomly the decision in which direction to move on.

$$X_t = X_{t-1} + e_t$$

where e_t is random with equal probability for -1 and 1 .

```
set.seed(42)
step<- 1 - 2 * round(runif(100))
X_t<-cumsum(step)
plot(X_t, type="lines")
```

Warning in plot.xy(xy, type, ...): plot type 'lines' will be truncated to first character

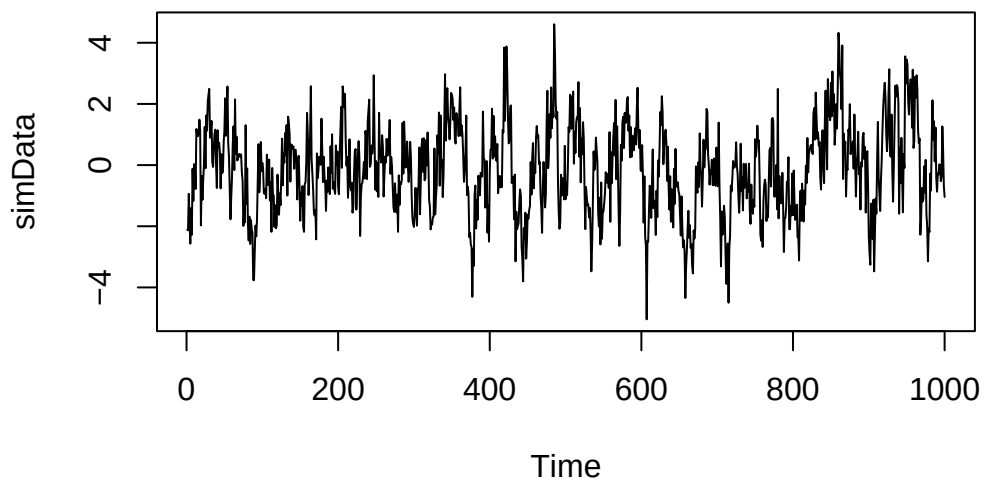


Simulation $AR(p)$ Processes

We would like to simulate:

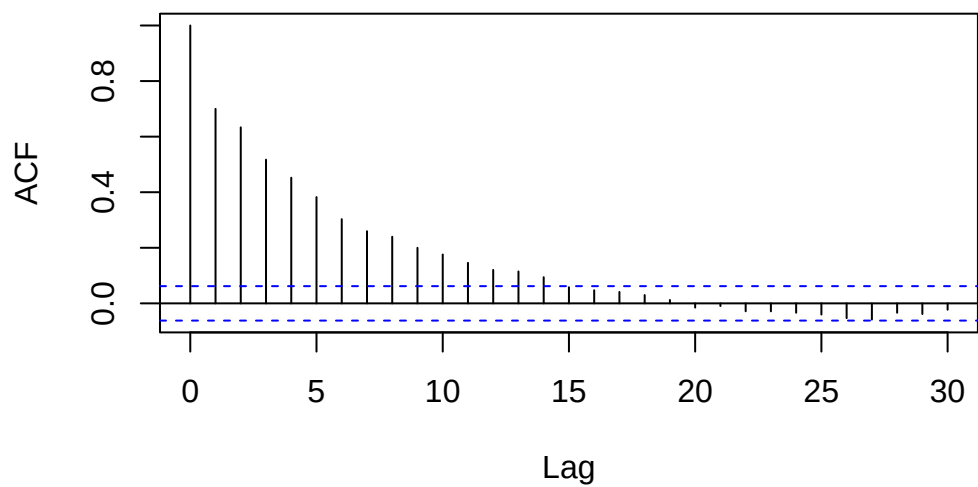
$$X_t = 0.5X_{t-1} + 0.3X_{t-2} + e_t \text{ with } e_t \sim N(0, 1) \text{ iid.}$$

```
set.seed(42)
ar.coefficients <- c(0.5, 0.3)
simData <- arima.sim(n=1000,model=list(ar=ar.coefficients))
plot(simData)
```

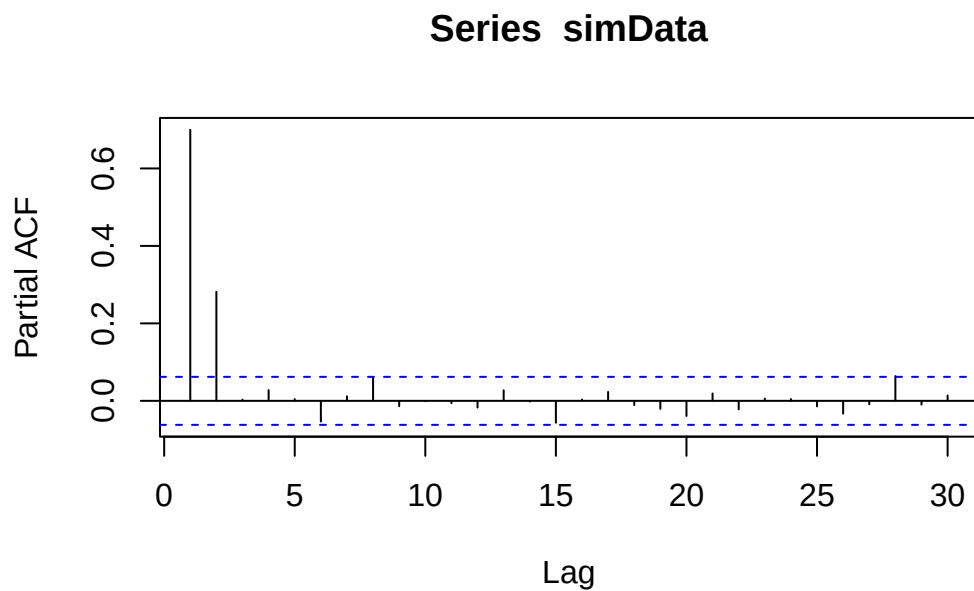


```
acf(simData)
```

Series simData



```
pacf(simData)
```



Stationarity of $AR(p)$ Models

$AR(p)$ models are stationary when the:

- mean is 0
- absolute value of the roots of the characteristic polynomial is all larger than 1

```
polyroot(c(1,-0.8,-0.4))
```

```
[1] 0.8708287+0i -2.8708287+0i
```

As the first value has an absolute value below 1, the time series is not stationary.

Key Properties of $AR(p)$ Models

The $AR(1)$ process:

$$X_t = 0.75 * X_t + e_t$$

is stationary because the characteristic polynomial:

```
polyroot(c(1,-0.75))
```

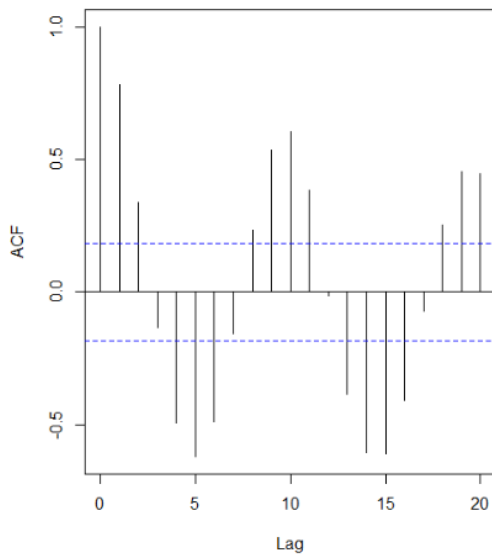
```
[1] 1.333333+0i
```

For the model identification of $AR(p)$ processes, two properties are key:

- Autocorrelation decays quickly or displays a damped sinusoid
- Partial autocorrelation function has a clear cut-off at p

Model Order Identification

Plug-in estimated ACF of $\log(\text{lynx})$



Partial ACF

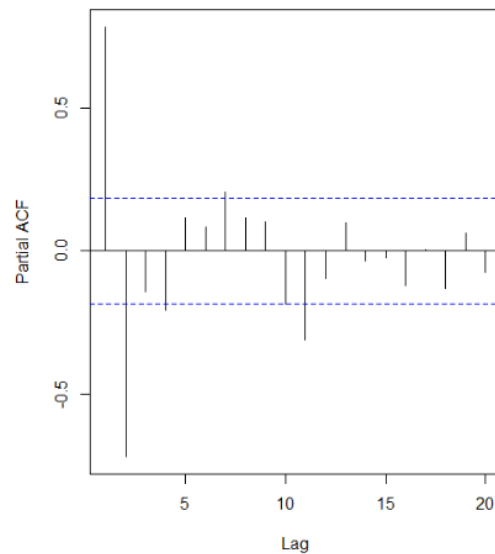


Figure 9: Plug-in estimated ACF and PACF

- ACF displays damped sinusoid
- PACF has clear cut-off after lag 11, 7, 4 or 2 (depending on the threshold).

Try fitting an $AR(2)$, $AR(4)$, $AR(7)$, $AR(11)$

W-05-Timeseries-AR-MA

Model Parameter Identification

Time series are often not centered: I.e. instead of a classical AR(p) process, a centered version has to be fitted.

$$Y_t = m + X_t = m + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t$$

Fitting parameters are: - the global mean m - the AR-coefficients $\alpha_1, \dots, \alpha_p$ - the parameters defining the distribution of the innovation e_t . Here, we assume a normal distribution.

Model Parameter from Ordinary Least Square (OLS)

We want to estimate the parameters from the model equation, we have a total of $n - p$ points for this, because we have a total of n time segments and each equation links p time instance.

The OLS procedure is:

1. Estimate the global mean $\hat{m} = \bar{y} = \frac{1}{n} \sum_{t=1}^n y_t$ and determine $x_t = y_t - \hat{m}$
2. Carry out a regression on x_t without intercept. The resulting parameters are $\hat{\alpha}_1, \dots, \hat{\alpha}_p$
3. Estimate the standard deviation of the innovation $\hat{\sigma}_E^2$ deviation of the residual.

```
ar.ols(data, order=p)
```

Yule-Walker Equations

Basic idea: The model coefficients and the ACF values are entangled. The values of the auto-correlation function depend on the model coefficients. A careful calculation for an AR(p) model yields the following equations, called Yule-Walker equations:

$$\sum_{i=1}^p \alpha_i \rho(k-i) = \rho(k) \quad \text{for } k > 0$$
$$\sum_{i=1}^p \alpha_i \rho(i) = \rho(0) - \sigma_E^2 \quad \text{for } k = 0$$

Note: Note that the autocorrelation function is symmetric, i.e. $\rho(-k) = \rho(k)$ for all k .

To fit the AR(p) model parameters, we must solve the above equations with the estimated values of the auto correlation function.

```
ar.yw(data,order=p)
```

Further options for fitting

Burg's algorithm

Offers another fitting cost function that exploit also the first p function values.

```
ar.burg(data,order.max=p)
```

Maximum likelihood estimator (MLE)

Determines the parameter based on an optimization of the maximum likelihood function

```
arima(Data,order=c(2,0,0))
```

Which method to use when?

One option is to use the Akaike information criterion (AIC), which estimates the relative amount of information lost by the considered model. The less information a model loses, the higher the quality of that model.

```
f.burg<-ar.burg(log(lynx))  
plot(  
  0:f.burg$order.max,  
  f.burg$aic,  
  type="l"  
)
```



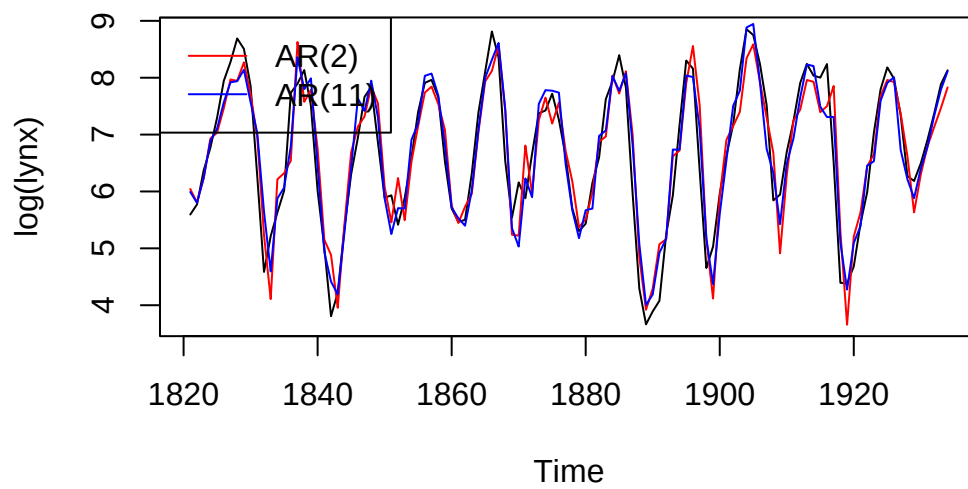
Note: No order parameter is given any more.

Model Diagnostic

Example

```
# Fitting the two models
fit.2<-arima(log(lynx),order=c(2,0,0))
fit.11<-arima(log(lynx),order=c(11,0,0))

# Display the time series and model predictions
plot(log(lynx))
lines(log(lynx) - fit.2$resid, col = "red")
lines(log(lynx) - fit.11$resid, col = "blue")
legend("topleft",
      legend = c("AR(2)", "AR(11)"),
      col = c("red", "blue"),
      lty = 1)
```



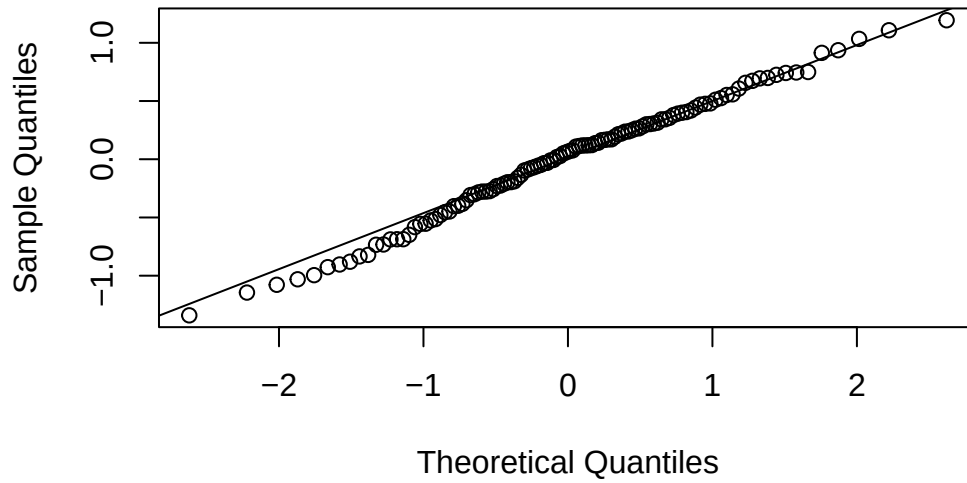
QQ-Plot

Compare quantile of measurement data with theoretical distribution

- X-Axis: Theoretical distribution
- Y-Axis: Actual data distribution

```
qqnorm(fit.2$residuals,main="Normal Q-Q Plot: AR(2)")  
qqline(fit.2$residuals)
```

Normal Q-Q Plot: AR(2)



Moving Average Models (MA)

A moving average MA(q) model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

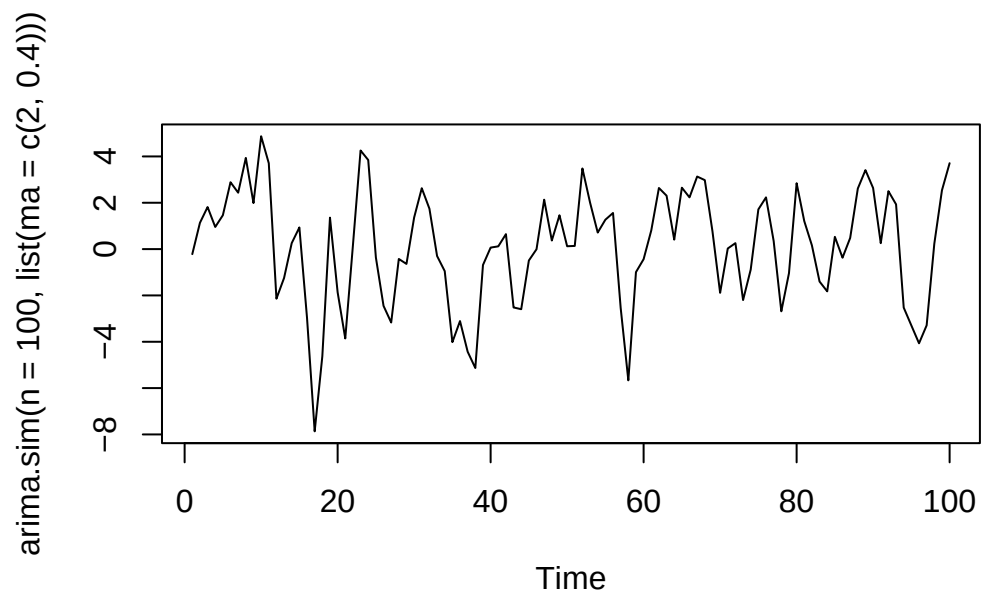
with the time series E_t of innovations.

i Note

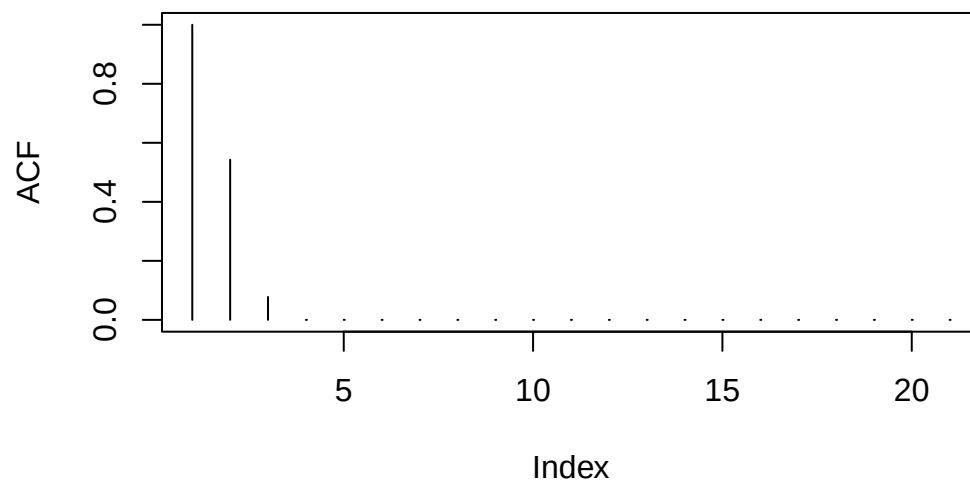
A moving average model does not have an explicit reference to previous values; only the innovation parts are considered.

```
set.seed(42)

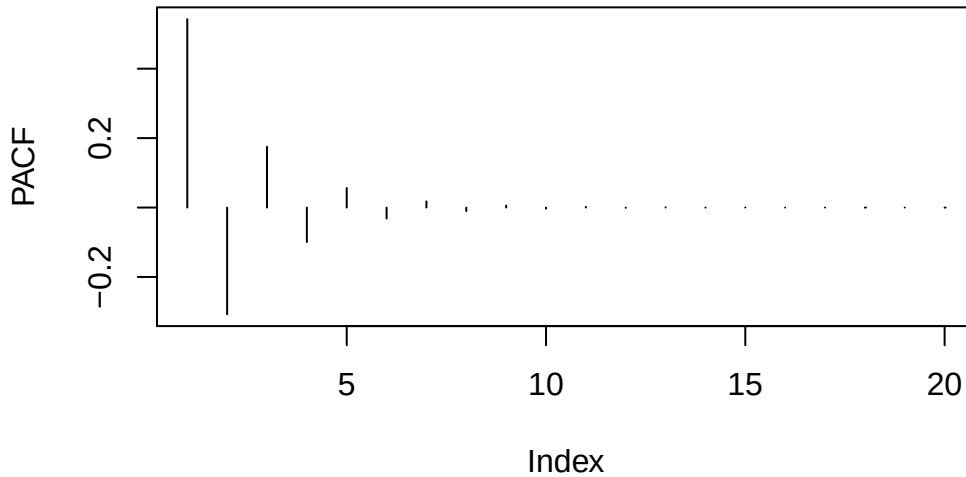
#MA(2)
plot(arima.sim(n=100,list(ma=c(2,0.4))))
```



```
plot(ARMAacf(ma=c(2,0.4),lag.max=20),type="h",ylab="ACF")
```



```
plot(ARMAacf(ma=c(2,0.4),pacf=T,lag.max=20),type="h",ylab="PACF")
```



Generalization: In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q .

W-06-Timeseries-Diagnostics-MA

Moving Average Model

A moving average $MA(q)$ model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

with the time series E_t of innovations. The ACF displays cutoff and PACF displays fast decay. In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q .

Invertibility

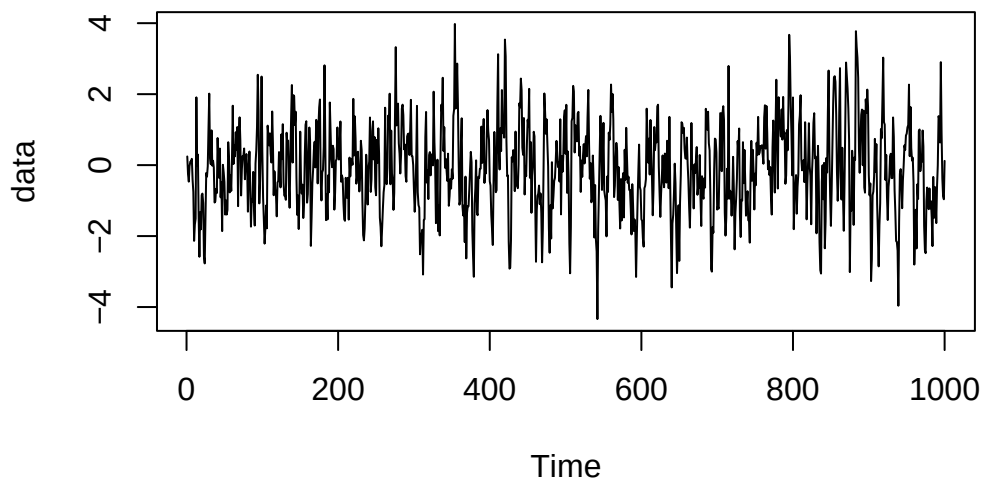
An $MA(q)$ model is invertible if the absolute value of the roots of the characteristic polynomial are larger than 1. Invertible $MA(q)$ processes can be expressed as $AR(\infty)$ processes.

Model Identification

$MA(q)$ models are (weakly) stationary, display clear cut-off in the ACF and PACF displays a fast decaying or a damped sinusoid behaviour.

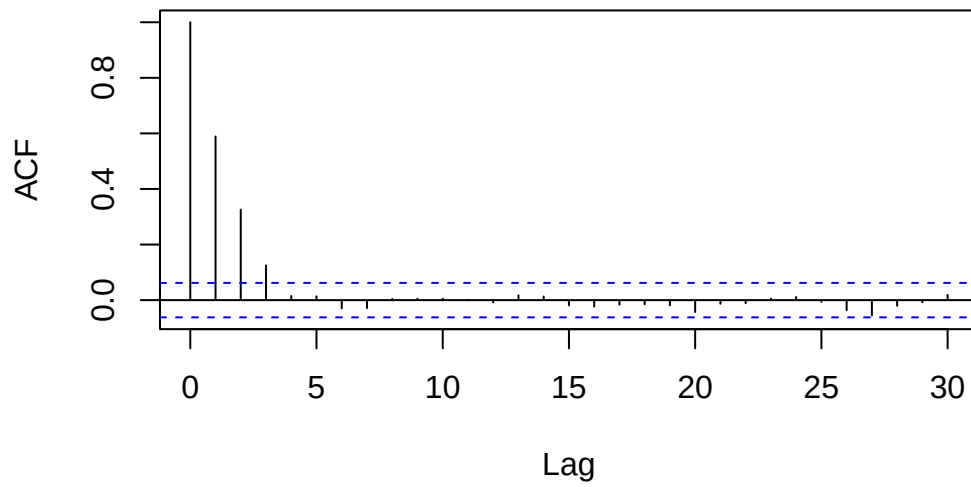
If these conditions are met, a $MA(q)$ model may be a good choice.

```
# Let us simulate an MA(3) process  
data<-arima.sim(n=1000,list(ma=c(0.6,0.4,0.2)))  
plot(data)
```



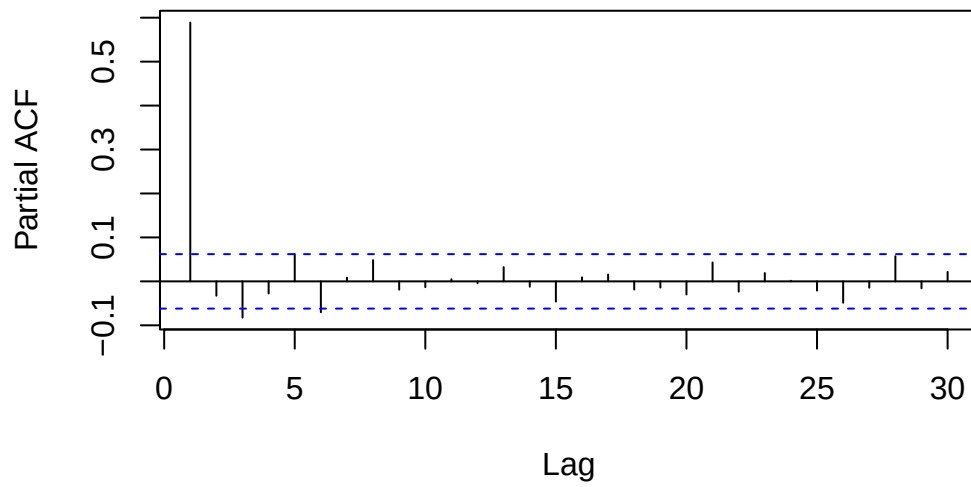
```
acf(data)
```

Series data



```
pacf(data)
```

Series data



Parameter Identification

The parameters of an $MA(q)$ model can be estimated with:

- Yule-Walker equations: As in the case of the $AR(p)$ processes, the parameters of $MA(q)$ processes can also be estimated by the Yule-Walker equations (after a global mean has been subtracted).
- Conditional sum of squares: `arima(data, order=c(...), method="CSS")`
- Maximum-likelihood estimation (default method in R): `arima(data, order=c(...), method="CSS-MLE")`

ARMA Models

An $ARMA(p, q)$ process combines an $AR(p)$ and a $MA(q)$ process

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + E_t + \beta_1 E_{t-1} + \dots + \beta_q E_{t-q}$$

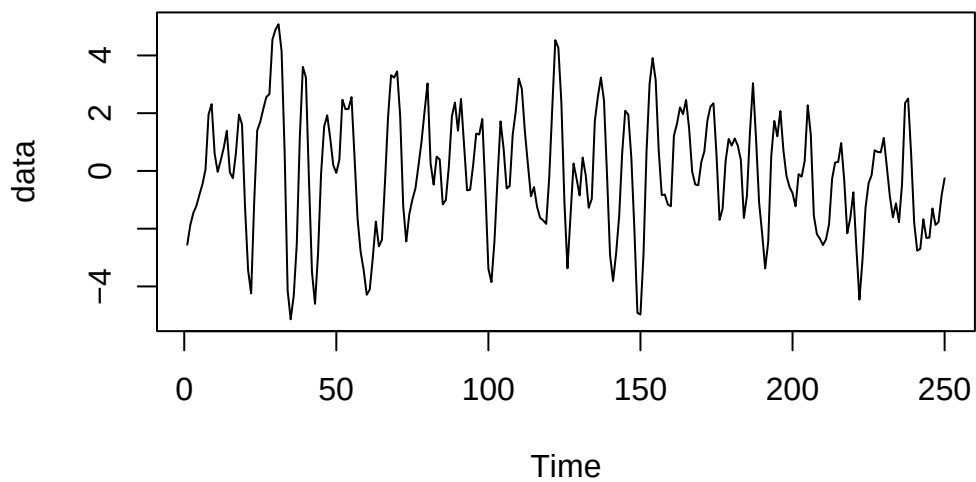
Definition

- Stationarity An $ARMA(p, q)$ model is stationary if the absolute value of the roots of characteristic polynomial is larger than 1. > Note: Analogy to the $AR(p)$ process
- Invertability An $ARMA(p, q)$ model is invertible if the absolute value of the roots of characteristic polynomial is larger than 1. > Note: Analogy to the $MA(q)$ process
- Consequence: Any stationary and invertible $ARMA(p, q)$ model can either be rewritten in the form of an $AR(\infty)$ or an $MA(\infty)$ model.

i Note

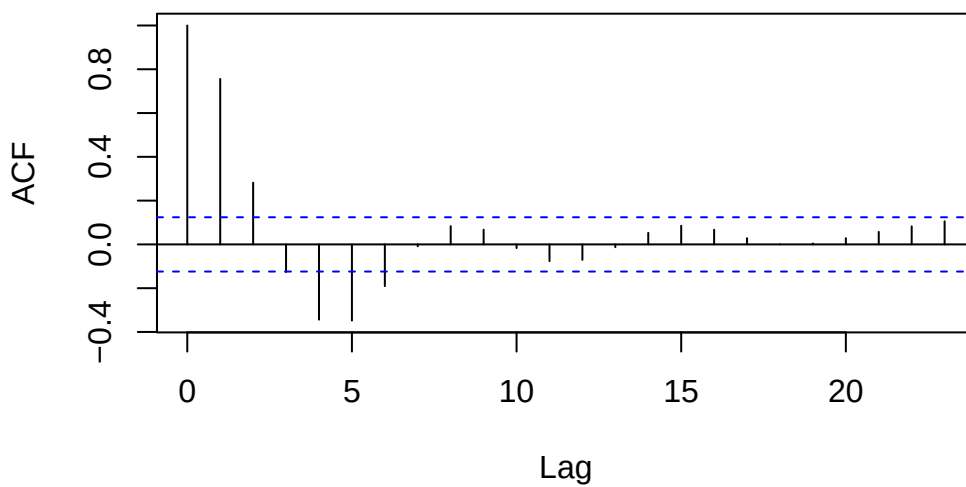
In practise $ARMA(p, q)$ models are preferred because they require fewer parameters.

```
data<-arima.sim(n=250,list(ar=c(1.1,-0.5),ma=c(0.4)))
plot(data)
```

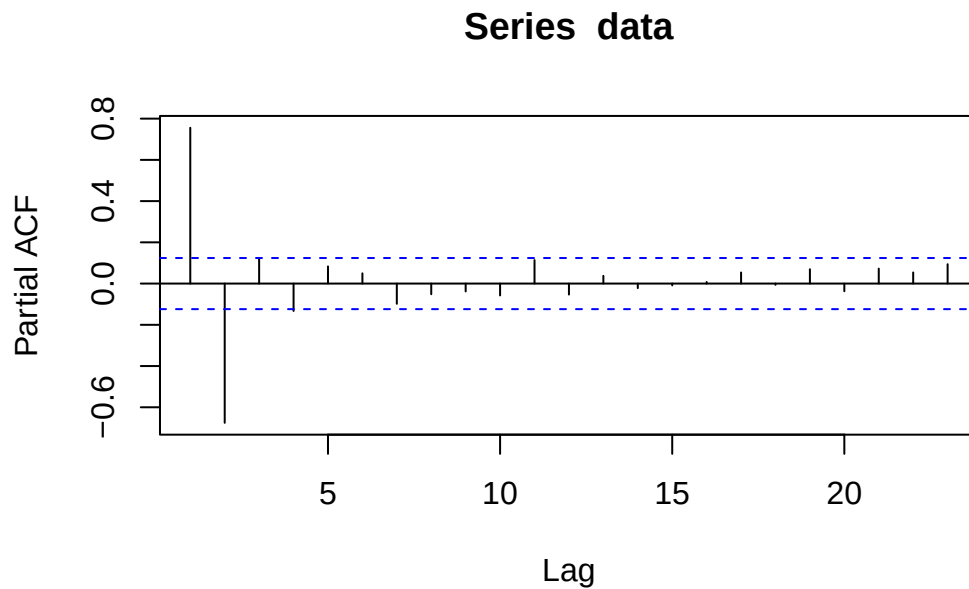


```
acf(data)
```

Series data



```
pacf(data)
```



Order Identification

Generic workflow for fitting:

- Is the model stationary?
- Do the properties of (P)ACF match?
- Which orders p, q could be cut off in the (P)ACF?

Note: Even with theoretical (P)ACF the identification of the cut-off can be challenging.

Fitting ARMA Models

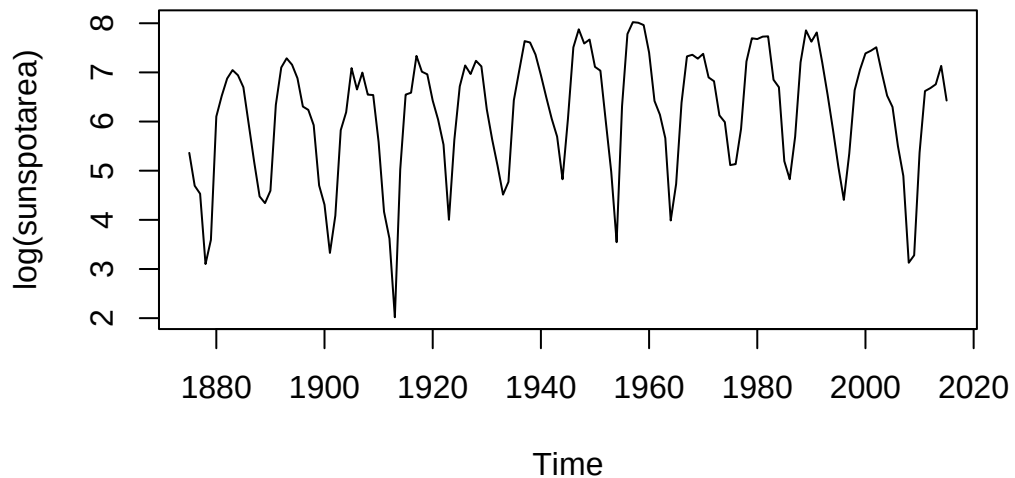
- Conditional sum of squares: Applicable for any invertible $ARMA(p, q)$ process
`arima(data, order=c(...), method=«CSS»)`
- Maximum likelihood estimation: Performs maximum likelihood estimation after CSS-process
`arima(..., order=c(...), method="CSS-ML")`

```
library(fpp2)
```

```
-- Attaching packages ----- fpp2 2.5.1 --
```

```
v ggplot2 4.0.1    v fma      2.5  
v forecast 9.0.2    v expsmooth 2.3
```

```
plot(log(sunspotarea))
```



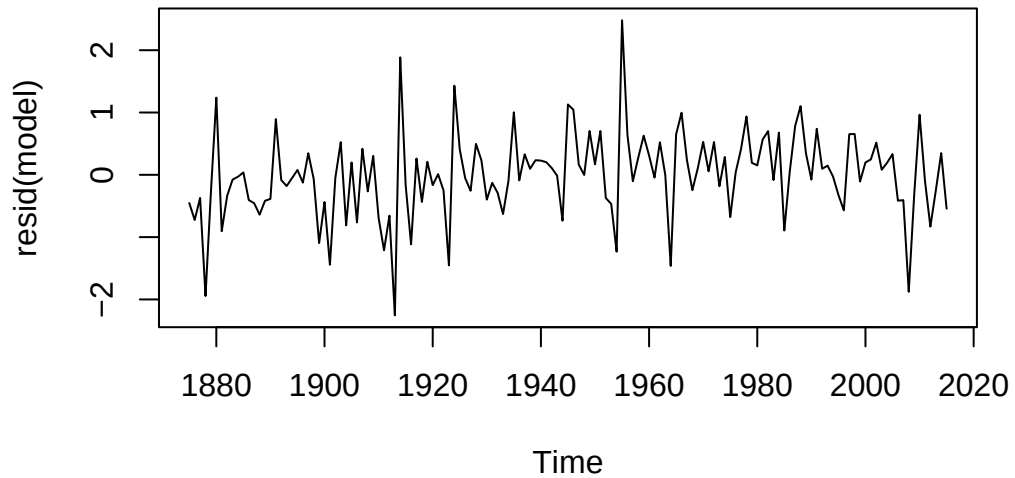
```
library(forecast)  
model <- auto.arima(log(sunspotarea), max.p = 10, max.q = 10, stationary = TRUE)  
model
```

```
Series: log(sunspotarea)  
ARIMA(2,0,1) with non-zero mean
```

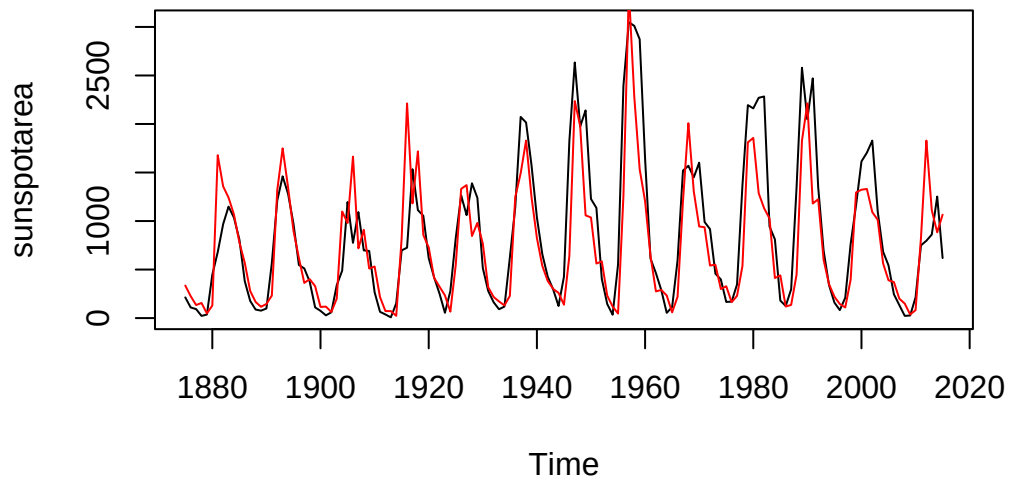
```
Coefficients:  
      ar1      ar2      ma1      mean
```

```
      1.4834  -0.7710  -0.5132  6.1885  
s.e.  0.0728   0.0605   0.0953  0.0983  
  
sigma^2 = 0.4794:  log likelihood = -147.06  
AIC=304.12   AICc=304.56   BIC=318.86
```

```
plot(resid(model))
```



```
plot(sunspotarea)  
lines(exp(log(sunspotarea)-resid(model)),col="red")
```



Advantages

Estimates all parameters above and provides error estimates.

Disadvantage

Works for any distribution of the innovation. For distributions that deviate strongly from Gaussian type, results may be unreliable.

Key Properties of Models

Property	AR(p)	MA(q)	ARMA(p,q)
ACF	Fast decay / damped sinusoid	Cut-off at q	Drop at q
PACF	Cut-off at p	Fast decay	Drop at p

Figure 10: Model Properties

W-07-Timeseries-ARMA-GLS

Linear Regression with timeseries

Principle

Model the target time series Y_t as a (linear) combination/function of the constituent, with an error term E_t and the realizations $x_{t1}, \dots, x_{tp}$ of the p time series. Major challenge: In real-world data, the errors can be strongly correlated.

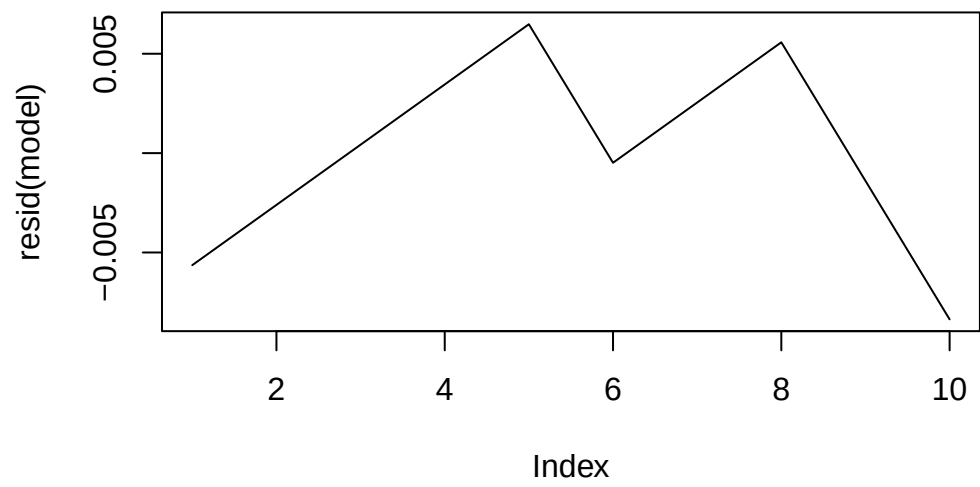
Find Model Coefficients

An assumption of linear regression is that the error E_t should be stationary and independent of the input time series. Often it is non-white noise and displays serial correlation. Perform a linear fit on the data and analyze the residuals:

```
# Sample Data
la_data <- data.frame(
  year = 2010:2019,
  value = c(3.79, 3.82, 3.85, 3.88, 3.91, 3.93, 3.96, 3.99, 4.01, 4.03)
)

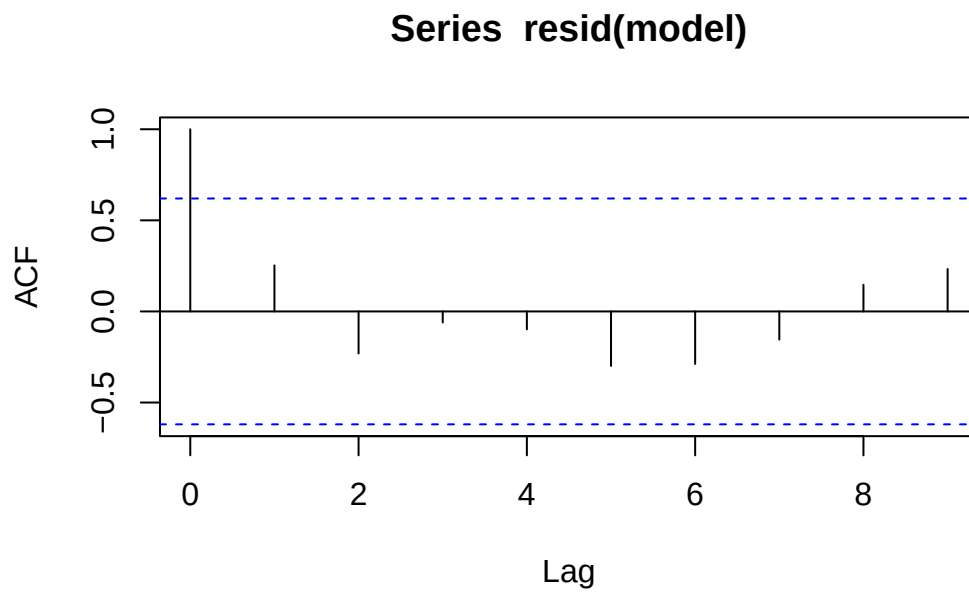
# Perform Linear Fit
model <- lm(value ~ year, data = la_data)

plot(resid(model), type="l")
```

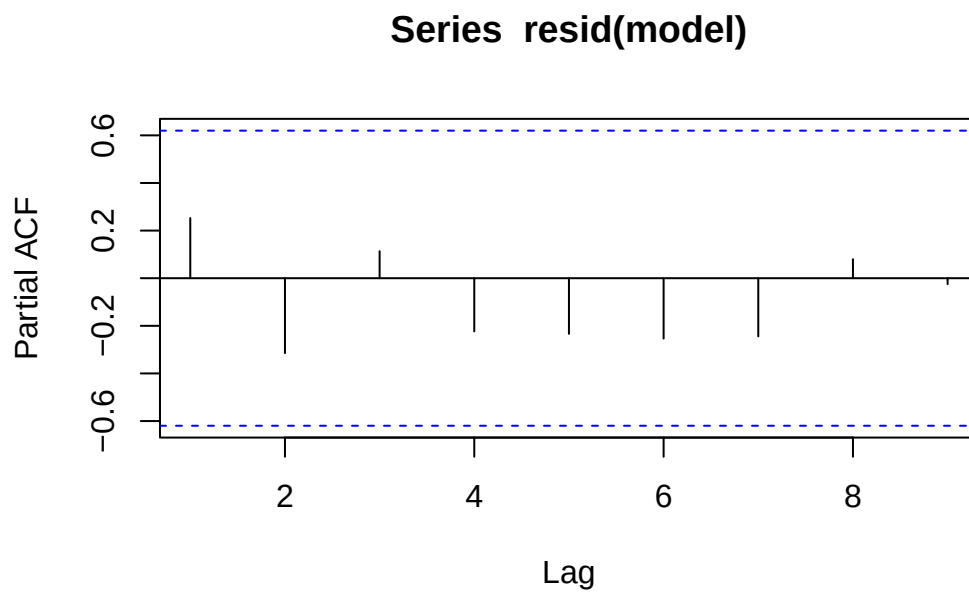


Is there a structure in the residuals?

```
acf(resid(model))
```

```
pacf(resid(model))
```



Error displays a serial connection and may be modeled by an AR(1) process. Whenever the error terms are dependent, the error estimation of the linear regression is invalid.

Note

Standard deviation is invalid if residuals are correlated

```
summary(model)
```

Call:

```
lm(formula = value ~ year, data = la_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0083636	-0.0023030	-0.0000303	0.0032273	0.0064848

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.041e+01	1.116e+00	-45.19	6.35e-11 ***
year	2.697e-02	5.538e-04	48.70	3.50e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.00503 on 8 degrees of freedom

Multiple R-squared: 0.9966, Adjusted R-squared: 0.9962

F-statistic: 2372 on 1 and 8 DF, p-value: 3.498e-11

Durbin-Watson Test

The Durbin-Watson (DW) test is a statistical test used to detect the presence of autocorrelation (specifically first-order serial correlation) in the residuals from a regression analysis.

In a standard linear regression, we assume that residuals are independent. If they are correlated (e.g., if the error at time t is related to the error at time $t - 1$), your standard errors may be underestimated, leading to misleadingly “significant” results.

```
library(lmtest)
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

```
as.Date, as.Date.numeric
```

```
dwtest(model)
```

Durbin-Watson test

```
data: model
```

```
DW = 0.99211, p-value = 0.009606
```

```
alternative hypothesis: true autocorrelation is greater than 0
```

Cochrane-Orcutt

The Cochrane-Orcutt method is an iterative procedure used in econometrics and statistics to estimate a linear regression model in the presence of autocorrelation (specifically AR(1) errors).

When the Durbin-Watson test indicates that residuals are correlated, the standard Ordinary Least Squares (OLS) assumptions are violated. Cochrane-Orcutt “cleanses” the model by transforming the variables to eliminate this correlation.

Steps

1. Perform ordinary least square fit on data.
2. Analyse (P)ACF and fit AR(1) model on residuals.
3. Transform data according to parameter of the residuals' AR(1) process.
4. Fit the transformed data to determine the parameters of the linear model.

Generalized Least Square Methode

Generalized Least Squares (GLS) is a powerful extension of Ordinary Least Squares (OLS) used to estimate unknown parameters in a linear regression model when the underlying assumptions about the error terms are violated. Specifically, it is used when there is heteroscedasticity (non-constant variance) or autocorrelation (correlated errors).

Impact of Missing Variables

Correlated errors are often caused by missing input variables. In a perfect world, these parameters would be included in the model. However, often it is impossible to provide the input.

W-08-Timeseries-GLS-ARIMA

An ARIMA model is basically an ARMA model that has been “stretched” to include a trend. Because it contains that trend (the unit root), it doesn’t settle back to a constant average, making it non-stationary.

Definition

ARIMA stands for AutoRegressive Integrated Moving Average. It is a powerful and flexible class of statistical models used for analyzing and forecasting time-series data. Unlike simple linear regression, ARIMA explicitly accounts for the dependencies between observations over time. The model is defined by three parameters p, d, q .

Interpreting the Model Output

```
library(TSA)
```

```
Registered S3 methods overwritten by 'TSA':
```

```
  method      from  
fitted.Arima forecast  
plot.Arima   forecast
```

```
Attaching package: 'TSA'
```

```
The following objects are masked from 'package:stats':
```

```
  acf, arima
```

```
The following object is masked from 'package:utils':
```

```
  tar
```

```
data(oil.price)
fit<-arima(log(oil.price),order=c(2,1,1))
fit
```

Call:

```
arima(x = log(oil.price), order = c(2, 1, 1))
```

Coefficients:

	ar1	ar2	ma1
	-0.3240	0.0150	0.5981
s.e.	0.2653	0.1107	0.2571

sigma^2 estimated as 0.006642: log likelihood = 261.12, aic = -516.25

This ARIMA(2,1,1) model is equal to an ARMA(3,1) model.

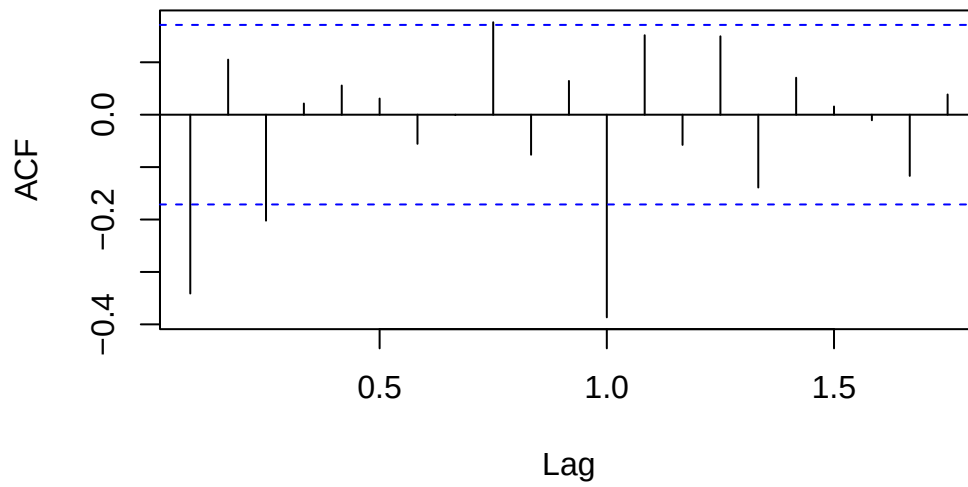
W-09-Timeseries-ARIMA-SARIMA

Sarima Models

SARIMA stands for **S**easonal **A**utoregressive **I**ntegrated **M**oving **A**verage. It is an extension of the standard ARIMA model specifically designed to handle univariate time series data that contains a seasonal component. While ARIMA focuses on the relationship between points in time (trend), SARIMA adds a layer to account for cycles that repeat over specific intervals (e.g., monthly sales spikes or daily temperature changes).

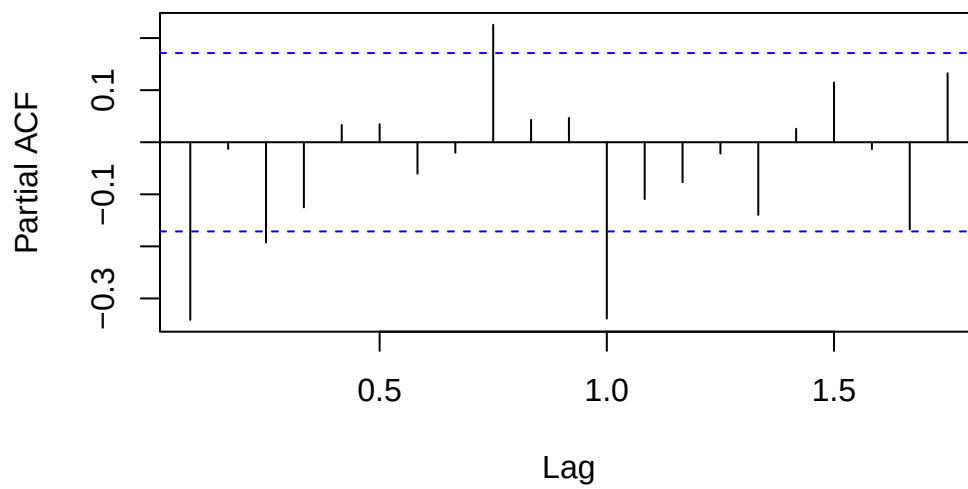
```
data("AirPassengers")
acf(diff(diff(log(AirPassengers),lag=12)))
```

Series $\text{diff}(\text{diff}(\log(\text{AirPassengers}), \text{lag} = 12))$



```
pacf(diff(diff(log(AirPassengers), lag=12)))
```

Series $\text{diff}(\text{diff}(\log(\text{AirPassengers}), \text{lag} = 12))$



```
fit<-arima(log(AirPassengers),order=c(0,1,1),seasonal=c(0,1,1))
fit
```

Call:

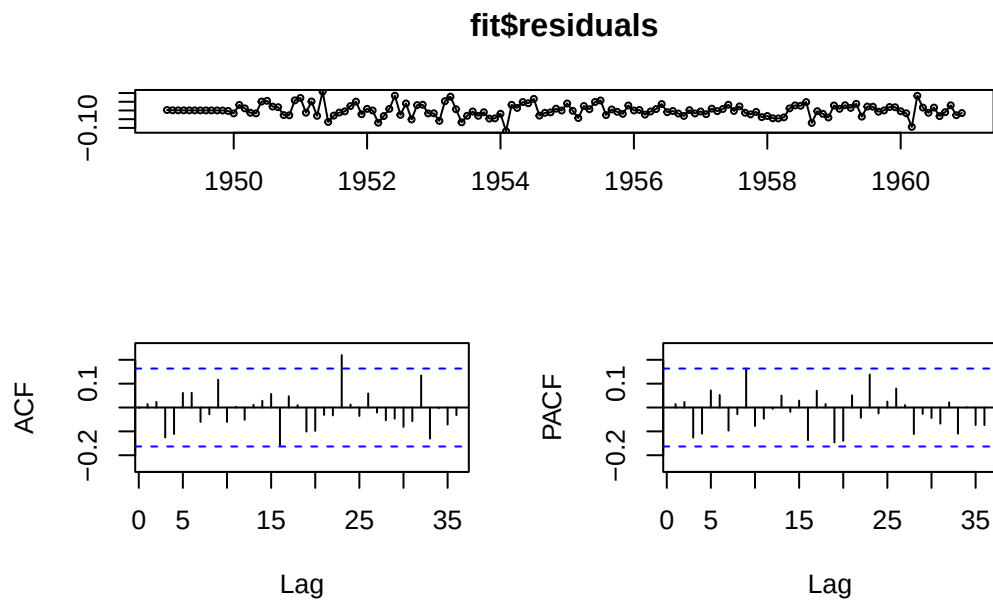
```
arima(x = log(AirPassengers), order = c(0, 1, 1), seasonal = c(0, 1, 1))
```

Coefficients:

	ma1	sma1
	-0.4018	-0.5569
s.e.	0.0896	0.0731

sigma² estimated as 0.001348: log likelihood = 244.7, aic = -485.4

```
library(forecast)
tsdisplay(fit$residuals)
```



```
fitAuto<-auto.arima(log(AirPassengers),ic="aic")
fitAuto
```

Series: log(AirPassengers)

ARIMA(0,1,1)(0,1,1)[12]

Coefficients:

	ma1	sma1
	-0.4018	-0.5569
s.e.	0.0896	0.0731

sigma^2 = 0.001371: log likelihood = 244.7
AIC=-483.4 AICc=-483.21 BIC=-474.77

Note: The automated procedure selects the same model.

How to fit SARIMA Models?

1. Estimate the seasonality period s visually. The order of differentiation D (for the seasonality part) is typically less or equal to 1.
2. Check if another differentiation of lag 1 is required for stationarity. If yes, try $d=1$, else $d=0$.
3. Analyse (P)ACF of the result of the two differentiations. Determine p,q from the short-lag behavior and P,Q from the multiple-of-the-period dependency.
4. Fit the model with `arima(data,order=c(p,d,q),seasonal=c(P,D,Q))`
5. Check the applicability of the model by analyzing the residuals (residuals should behave like (Gaussian) white noise).

Non-Linear Models

Linear models are models that can be formulated as linear combination of different time series elements X_t and errors/innovations E_t .

Note: An example are data with changing standard deviation.

Heteroscedasticity

Heteroscedasticity occurs when the variance of the residuals (errors) in a model is not constant across all levels of the independent variables or across time.

Note: Heteroskedastic time series are not stationary, because the variance is not equal for all time points.

ARCH Models

A time series is autoregressive conditional heteroskedastic of the order p (abbreviated ARCH(p)) if it has the form:

$$X_t = \sigma_t W_t \quad \text{with} \quad \sigma_t = \sqrt{\alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2}$$

Fitting ARCH Models

```
#load SMI data
data("EuStockMarkets")

esm <- EuStockMarkets
tmp <- EuStockMarkets[,2]
smi <- ts(tmp, start=start(esm), freq=frequency(esm))
lret.smi <- diff(log(smi))

#perform fit
library(tseries)
```

Registered S3 method overwritten by 'quantmod':

```
method      from
as.zoo.data.frame zoo
```

```
fit<-garch(lret.smi, order=c(0,2))
```

***** ESTIMATION WITH ANALYTICAL GRADIENT *****

I	INITIAL X(I)	D(I)
1	7.700685e-05	1.000e+00
2	5.000000e-02	1.000e+00
3	5.000000e-02	1.000e+00

IT	NF	F	RELDF	PRELDF	RELDX	STPPAR	D*STEP	NPRELDF
0	1	-7.796e+03						
1	8	-7.797e+03	7.76e-05	1.39e-04	2.7e-05	1.4e+11	2.7e-06	9.94e+06

2	9	-7.797e+03	7.84e-07	8.41e-07	2.1e-05	2.0e+00	2.7e-06	3.10e+00
3	17	-7.803e+03	8.48e-04	1.27e-03	2.6e-01	2.0e+00	4.5e-02	3.10e+00
4	19	-7.803e+03	5.19e-06	7.74e-06	2.0e-02	2.0e+00	4.5e-03	1.44e-03
5	20	-7.803e+03	2.15e-07	2.70e-06	2.0e-02	2.0e+00	4.5e-03	3.72e-04
6	21	-7.803e+03	3.22e-06	2.03e-05	1.3e-02	1.9e+00	2.2e-03	3.91e-04
7	24	-7.805e+03	1.73e-04	1.62e-04	9.7e-02	4.9e-01	2.1e-02	1.87e-04
8	25	-7.805e+03	9.03e-05	1.05e-04	1.6e-01	0.0e+00	3.8e-02	1.05e-04
9	26	-7.805e+03	1.19e-05	1.35e-05	4.7e-02	0.0e+00	1.3e-02	1.35e-05
10	27	-7.805e+03	1.16e-07	1.07e-07	4.1e-03	0.0e+00	1.1e-03	1.07e-07
11	28	-7.805e+03	1.13e-09	1.10e-09	3.2e-04	0.0e+00	1.0e-04	1.10e-09
12	29	-7.805e+03	8.85e-12	7.83e-12	4.0e-05	0.0e+00	1.1e-05	7.83e-12

***** RELATIVE FUNCTION CONVERGENCE *****

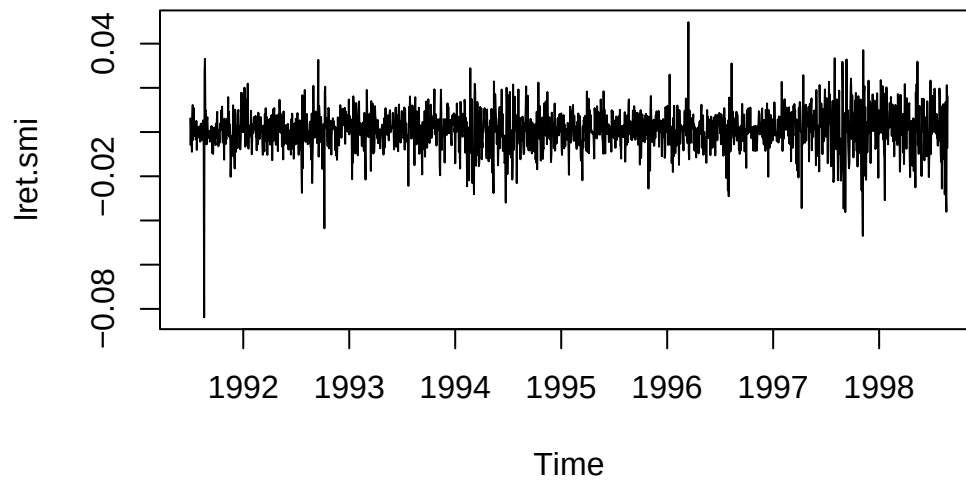
FUNCTION	-7.805450e+03	RELDX	3.982e-05
FUNC. EVALS	29	GRAD. EVALS	13
PRELDF	7.826e-12	NPRELDF	7.826e-12

I	FINAL X(I)	D(I)	G(I)
1	6.567582e-05	1.000e+00	1.512e+00
2	1.309070e-01	1.000e+00	1.394e-03
3	1.074336e-01	1.000e+00	-8.848e-04

```
fit$coef
```

	a0	a1	a2
	6.567582e-05	1.309070e-01	1.074336e-01

```
plot(lret.smi)
```



```
fit$fitted.values
```

Time Series:

Start = c(1991, 131)

End = c(1998, 169)

Frequency = 260

	sigt	-sigt
1991.500	NA	NA
1991.504	NA	NA
1991.508	0.008619948	-0.008619948
1991.512	0.008413776	-0.008413776
1991.515	0.008192281	-0.008192281
1991.519	0.008738385	-0.008738385
1991.523	0.008954976	-0.008954976
1991.527	0.009492994	-0.009492994
1991.531	0.009133273	-0.009133273
1991.535	0.009107082	-0.009107082
1991.538	0.009010984	-0.009010984
1991.542	0.008347955	-0.008347955
1991.546	0.008228813	-0.008228813
1991.550	0.008218464	-0.008218464
1991.554	0.008182066	-0.008182066

1991.558 0.008192224 -0.008192224
1991.562 0.008379568 -0.008379568
1991.565 0.008776044 -0.008776044
1991.569 0.008527705 -0.008527705
1991.573 0.008162093 -0.008162093
1991.577 0.008164387 -0.008164387
1991.581 0.008160428 -0.008160428
1991.585 0.008151230 -0.008151230
1991.588 0.008112088 -0.008112088
1991.592 0.008156176 -0.008156176
1991.596 0.008296850 -0.008296850
1991.600 0.008366156 -0.008366156
1991.604 0.008332257 -0.008332257
1991.608 0.008198592 -0.008198592
1991.612 0.008183117 -0.008183117
1991.615 0.008274863 -0.008274863
1991.619 0.008207454 -0.008207454
1991.623 0.008337280 -0.008337280
1991.627 0.008361690 -0.008361690
1991.631 0.008594171 -0.008594171
1991.635 0.031486098 -0.031486098
1991.638 0.030248146 -0.030248146
1991.642 0.016931529 -0.016931529
1991.646 0.013959892 -0.013959892
1991.650 0.009355178 -0.009355178
1991.654 0.008882279 -0.008882279
1991.658 0.008496063 -0.008496063
1991.662 0.008529784 -0.008529784
1991.665 0.008638284 -0.008638284
1991.669 0.008397910 -0.008397910
1991.673 0.008152679 -0.008152679
1991.677 0.008146693 -0.008146693
1991.681 0.008157023 -0.008157023
1991.685 0.008147135 -0.008147135
1991.688 0.008261294 -0.008261294
1991.692 0.008293718 -0.008293718
1991.696 0.008277651 -0.008277651
1991.700 0.008255187 -0.008255187
1991.704 0.008146510 -0.008146510
1991.708 0.008159992 -0.008159992
1991.712 0.008849384 -0.008849384
1991.715 0.008685582 -0.008685582
1991.719 0.008863515 -0.008863515

1991.723 0.008746134 -0.008746134
1991.727 0.008149004 -0.008149004
1991.731 0.008242323 -0.008242323
1991.735 0.008250394 -0.008250394
1991.738 0.008288147 -0.008288147
1991.742 0.008489278 -0.008489278
1991.746 0.008437827 -0.008437827
1991.750 0.008255917 -0.008255917
1991.754 0.008319251 -0.008319251
1991.758 0.008882839 -0.008882839
1991.762 0.008645265 -0.008645265
1991.765 0.008106061 -0.008106061
1991.769 0.008430871 -0.008430871
1991.773 0.008380250 -0.008380250
1991.777 0.008160663 -0.008160663
1991.781 0.008337303 -0.008337303
1991.785 0.008374722 -0.008374722
1991.788 0.008201513 -0.008201513
1991.792 0.008148414 -0.008148414
1991.796 0.008867868 -0.008867868
1991.800 0.008749505 -0.008749505
1991.804 0.009138825 -0.009138825
1991.808 0.008955490 -0.008955490
1991.812 0.008305139 -0.008305139
1991.815 0.008359014 -0.008359014
1991.819 0.008197059 -0.008197059
1991.823 0.009280959 -0.009280959
1991.827 0.009340271 -0.009340271
1991.831 0.008348872 -0.008348872
1991.835 0.008243151 -0.008243151
1991.838 0.008444371 -0.008444371
1991.842 0.008355787 -0.008355787
1991.846 0.008235493 -0.008235493
1991.850 0.008449480 -0.008449480
1991.854 0.008391644 -0.008391644
1991.858 0.008235631 -0.008235631
1991.862 0.009977428 -0.009977428
1991.865 0.009640597 -0.009640597
1991.869 0.008765698 -0.008765698
1991.873 0.008673617 -0.008673617
1991.877 0.008230351 -0.008230351
1991.881 0.008186538 -0.008186538
1991.885 0.010934059 -0.010934059

1991.888 0.011379417 -0.011379417
1991.892 0.009069809 -0.009069809
1991.896 0.008210180 -0.008210180
1991.900 0.009055160 -0.009055160
1991.904 0.009617014 -0.009617014
1991.908 0.009828552 -0.009828552
1991.912 0.009456009 -0.009456009
1991.915 0.008531402 -0.008531402
1991.919 0.008880995 -0.008880995
1991.923 0.010566194 -0.010566194
1991.927 0.010407740 -0.010407740
1991.931 0.009203200 -0.009203200
1991.935 0.009094073 -0.009094073
1991.938 0.008664659 -0.008664659
1991.942 0.008194982 -0.008194982
1991.946 0.008324782 -0.008324782
1991.950 0.008659282 -0.008659282
1991.954 0.008997751 -0.008997751
1991.958 0.009314972 -0.009314972
1991.962 0.008750371 -0.008750371
1991.965 0.008158372 -0.008158372
1991.969 0.008148114 -0.008148114
1991.973 0.008572321 -0.008572321
1991.977 0.008949577 -0.008949577
1991.981 0.008548992 -0.008548992
1991.985 0.010383200 -0.010383200
1991.988 0.009985221 -0.009985221
1991.992 0.008104062 -0.008104062
1991.996 0.010281157 -0.010281157
1992.000 0.010408724 -0.010408724
1992.004 0.008586744 -0.008586744
1992.008 0.008104062 -0.008104062
1992.012 0.008104062 -0.008104062
1992.015 0.010887462 -0.010887462
1992.019 0.010573012 -0.010573012
1992.023 0.008582480 -0.008582480
1992.027 0.008667570 -0.008667570
1992.031 0.009899865 -0.009899865
1992.035 0.009587678 -0.009587678
1992.038 0.008294730 -0.008294730
1992.042 0.008727087 -0.008727087
1992.046 0.011685818 -0.011685818
1992.050 0.011124312 -0.011124312

1992.054 0.008852072 -0.008852072
1992.058 0.008488556 -0.008488556
1992.062 0.008191479 -0.008191479
1992.065 0.008161264 -0.008161264
1992.069 0.008135442 -0.008135442
1992.073 0.008133394 -0.008133394
1992.077 0.008819991 -0.008819991
1992.081 0.008700958 -0.008700958
1992.085 0.008882336 -0.008882336
1992.088 0.008763225 -0.008763225
1992.092 0.008229312 -0.008229312
1992.096 0.008193479 -0.008193479
1992.100 0.008105145 -0.008105145
1992.104 0.008459825 -0.008459825
1992.108 0.008753527 -0.008753527
1992.112 0.008423248 -0.008423248
1992.115 0.008202114 -0.008202114
1992.119 0.008591399 -0.008591399
1992.123 0.008491258 -0.008491258
1992.127 0.008141547 -0.008141547
1992.131 0.008642189 -0.008642189
1992.135 0.008743048 -0.008743048
1992.138 0.008328796 -0.008328796
1992.142 0.008387007 -0.008387007
1992.146 0.008885047 -0.008885047
1992.150 0.008633258 -0.008633258
1992.154 0.008492690 -0.008492690
1992.158 0.008476420 -0.008476420
1992.162 0.008479088 -0.008479088
1992.165 0.008453904 -0.008453904
1992.169 0.008540355 -0.008540355
1992.173 0.008402056 -0.008402056
1992.177 0.008168542 -0.008168542
1992.181 0.008188718 -0.008188718
1992.185 0.008412027 -0.008412027
1992.188 0.009127916 -0.009127916
1992.192 0.008832324 -0.008832324
1992.196 0.008224585 -0.008224585
1992.200 0.008595076 -0.008595076
1992.204 0.009032970 -0.009032970
1992.208 0.009094936 -0.009094936
1992.212 0.008672668 -0.008672668
1992.215 0.009097053 -0.009097053

1992.219 0.008853036 -0.008853036
1992.223 0.008501397 -0.008501397
1992.227 0.008456418 -0.008456418
1992.231 0.008951860 -0.008951860
1992.235 0.008862216 -0.008862216
1992.238 0.008259244 -0.008259244
1992.242 0.008247010 -0.008247010
1992.246 0.008276468 -0.008276468
1992.250 0.008455216 -0.008455216
1992.254 0.008327811 -0.008327811
1992.258 0.008372782 -0.008372782
1992.262 0.008612999 -0.008612999
1992.265 0.008534986 -0.008534986
1992.269 0.008707346 -0.008707346
1992.273 0.008603702 -0.008603702
1992.277 0.009412799 -0.009412799
1992.281 0.009123109 -0.009123109
1992.285 0.008924027 -0.008924027
1992.288 0.008899103 -0.008899103
1992.292 0.008616227 -0.008616227
1992.296 0.009315199 -0.009315199
1992.300 0.008871950 -0.008871950
1992.304 0.008118866 -0.008118866
1992.308 0.008104062 -0.008104062
1992.312 0.008370274 -0.008370274
1992.315 0.008766013 -0.008766013
1992.319 0.008542232 -0.008542232
1992.323 0.008224820 -0.008224820
1992.327 0.008160564 -0.008160564
1992.331 0.008685029 -0.008685029
1992.335 0.009043074 -0.009043074
1992.338 0.008580813 -0.008580813
1992.342 0.008169280 -0.008169280
1992.346 0.009668413 -0.009668413
1992.350 0.009713235 -0.009713235
1992.354 0.008967195 -0.008967195
1992.358 0.008592689 -0.008592689
1992.362 0.008421952 -0.008421952
1992.365 0.008790443 -0.008790443
1992.369 0.008794637 -0.008794637
1992.373 0.008401973 -0.008401973
1992.377 0.008268650 -0.008268650
1992.381 0.009366450 -0.009366450

1992.385 0.009668749 -0.009668749
1992.388 0.008906964 -0.008906964
1992.392 0.008380959 -0.008380959
1992.396 0.008282566 -0.008282566
1992.400 0.008285744 -0.008285744
1992.404 0.008187293 -0.008187293
1992.408 0.008158276 -0.008158276
1992.412 0.008363988 -0.008363988
1992.415 0.008292752 -0.008292752
1992.419 0.008193552 -0.008193552
1992.423 0.008193358 -0.008193358
1992.427 0.008225888 -0.008225888
1992.431 0.008678429 -0.008678429
1992.435 0.008514259 -0.008514259
1992.438 0.008130394 -0.008130394
1992.442 0.008121407 -0.008121407
1992.446 0.008534298 -0.008534298
1992.450 0.008749110 -0.008749110
1992.454 0.009135282 -0.009135282
1992.458 0.009027266 -0.009027266
1992.462 0.008773230 -0.008773230
1992.465 0.008996031 -0.008996031
1992.469 0.008696120 -0.008696120
1992.473 0.008481860 -0.008481860
1992.477 0.008418704 -0.008418704
1992.481 0.009280515 -0.009280515
1992.485 0.009016161 -0.009016161
1992.488 0.008286290 -0.008286290
1992.492 0.008384278 -0.008384278
1992.496 0.008459077 -0.008459077
1992.500 0.008590610 -0.008590610
1992.504 0.008494179 -0.008494179
1992.508 0.008440185 -0.008440185
1992.512 0.008470326 -0.008470326
1992.515 0.008831413 -0.008831413
1992.519 0.008714892 -0.008714892
1992.523 0.008304402 -0.008304402
1992.527 0.008980297 -0.008980297
1992.531 0.008957568 -0.008957568
1992.535 0.008483193 -0.008483193
1992.538 0.008772586 -0.008772586
1992.542 0.008617210 -0.008617210
1992.546 0.008400198 -0.008400198

1992.550 0.008345549 -0.008345549
1992.554 0.010504853 -0.010504853
1992.558 0.014164932 -0.014164932
1992.562 0.013539115 -0.013539115
1992.565 0.011425995 -0.011425995
1992.569 0.009789679 -0.009789679
1992.573 0.008265862 -0.008265862
1992.577 0.008247709 -0.008247709
1992.581 0.008836873 -0.008836873
1992.585 0.011073189 -0.011073189
1992.588 0.010314735 -0.010314735
1992.592 0.008347757 -0.008347757
1992.596 0.008258836 -0.008258836
1992.600 0.008394668 -0.008394668
1992.604 0.008841623 -0.008841623
1992.608 0.008541241 -0.008541241
1992.612 0.008346695 -0.008346695
1992.615 0.008527610 -0.008527610
1992.619 0.008293720 -0.008293720
1992.623 0.008226053 -0.008226053
1992.627 0.008382206 -0.008382206
1992.631 0.008369583 -0.008369583
1992.635 0.008379001 -0.008379001
1992.638 0.008576308 -0.008576308
1992.642 0.008776360 -0.008776360
1992.646 0.009592954 -0.009592954
1992.650 0.009581430 -0.009581430
1992.654 0.011947651 -0.011947651
1992.658 0.011731250 -0.011731250
1992.662 0.008886969 -0.008886969
1992.665 0.011125451 -0.011125451
1992.669 0.010664042 -0.010664042
1992.673 0.008254487 -0.008254487
1992.677 0.008232169 -0.008232169
1992.681 0.008429374 -0.008429374
1992.685 0.008990090 -0.008990090
1992.688 0.008656942 -0.008656942
1992.692 0.008581836 -0.008581836
1992.696 0.008498293 -0.008498293
1992.700 0.008302922 -0.008302922
1992.704 0.008720904 -0.008720904
1992.708 0.009319477 -0.009319477
1992.712 0.014739146 -0.014739146

1992.715 0.014008667 -0.014008667
1992.719 0.009860250 -0.009860250
1992.723 0.009506437 -0.009506437
1992.727 0.008713155 -0.008713155
1992.731 0.008385131 -0.008385131
1992.735 0.008535340 -0.008535340
1992.738 0.008705591 -0.008705591
1992.742 0.008432543 -0.008432543
1992.746 0.008128038 -0.008128038
1992.750 0.010104592 -0.010104592
1992.754 0.009776934 -0.009776934
1992.758 0.008114188 -0.008114188
1992.762 0.008218622 -0.008218622
1992.765 0.008867775 -0.008867775
1992.769 0.017985562 -0.017985562
1992.773 0.017662763 -0.017662763
1992.777 0.012471352 -0.012471352
1992.781 0.010785489 -0.010785489
1992.785 0.008523290 -0.008523290
1992.788 0.008248524 -0.008248524
1992.792 0.008426072 -0.008426072
1992.796 0.008388156 -0.008388156
1992.800 0.008452662 -0.008452662
1992.804 0.008467533 -0.008467533
1992.808 0.009092335 -0.009092335
1992.812 0.009491347 -0.009491347
1992.815 0.009076014 -0.009076014
1992.819 0.008478275 -0.008478275
1992.823 0.008323330 -0.008323330
1992.827 0.008470494 -0.008470494
1992.831 0.008289603 -0.008289603
1992.835 0.008914385 -0.008914385
1992.838 0.008774257 -0.008774257
1992.842 0.008174215 -0.008174215
1992.846 0.008390673 -0.008390673
1992.850 0.008600841 -0.008600841
1992.854 0.008502898 -0.008502898
1992.858 0.008414984 -0.008414984
1992.862 0.008296943 -0.008296943
1992.865 0.008140713 -0.008140713
1992.869 0.008159364 -0.008159364
1992.873 0.008680069 -0.008680069
1992.877 0.008581300 -0.008581300

1992.881 0.008987212 -0.008987212
1992.885 0.009332344 -0.009332344
1992.888 0.009699941 -0.009699941
1992.892 0.009181578 -0.009181578
1992.896 0.008535521 -0.008535521
1992.900 0.008407299 -0.008407299
1992.904 0.008207077 -0.008207077
1992.908 0.008466731 -0.008466731
1992.912 0.008447406 -0.008447406
1992.915 0.008235754 -0.008235754
1992.919 0.009258261 -0.009258261
1992.923 0.009560322 -0.009560322
1992.927 0.008592030 -0.008592030
1992.931 0.008163199 -0.008163199
1992.935 0.008162598 -0.008162598
1992.938 0.008254327 -0.008254327
1992.942 0.009050846 -0.009050846
1992.946 0.008892114 -0.008892114
1992.950 0.008859921 -0.008859921
1992.954 0.008761602 -0.008761602
1992.958 0.008258458 -0.008258458
1992.962 0.009078417 -0.009078417
1992.965 0.009073207 -0.009073207
1992.969 0.008957826 -0.008957826
1992.973 0.008706757 -0.008706757
1992.977 0.008643926 -0.008643926
1992.981 0.008524499 -0.008524499
1992.985 0.008355237 -0.008355237
1992.988 0.008345604 -0.008345604
1992.992 0.008134260 -0.008134260
1992.996 0.008104062 -0.008104062
1993.000 0.009700044 -0.009700044
1993.004 0.009441428 -0.009441428
1993.008 0.008529857 -0.008529857
1993.012 0.008449311 -0.008449311
1993.015 0.008104062 -0.008104062
1993.019 0.008373635 -0.008373635
1993.023 0.008516077 -0.008516077
1993.027 0.008286046 -0.008286046
1993.031 0.011199031 -0.011199031
1993.035 0.010734299 -0.010734299
1993.038 0.009557172 -0.009557172
1993.042 0.010978980 -0.010978980

1993.046 0.009701841 -0.009701841
1993.050 0.009437340 -0.009437340
1993.054 0.009218277 -0.009218277
1993.058 0.008652824 -0.008652824
1993.062 0.009236577 -0.009236577
1993.065 0.008706968 -0.008706968
1993.069 0.009390342 -0.009390342
1993.073 0.009441799 -0.009441799
1993.077 0.008850437 -0.008850437
1993.081 0.008524727 -0.008524727
1993.085 0.008951279 -0.008951279
1993.088 0.008932930 -0.008932930
1993.092 0.008811822 -0.008811822
1993.096 0.009957580 -0.009957580
1993.100 0.009291867 -0.009291867
1993.104 0.008118211 -0.008118211
1993.108 0.008182087 -0.008182087
1993.112 0.008285738 -0.008285738
1993.115 0.008249359 -0.008249359
1993.119 0.008146038 -0.008146038
1993.123 0.008424968 -0.008424968
1993.127 0.008427021 -0.008427021
1993.131 0.008270769 -0.008270769
1993.135 0.008346327 -0.008346327
1993.138 0.008553301 -0.008553301
1993.142 0.008926196 -0.008926196
1993.146 0.009128769 -0.009128769
1993.150 0.008890358 -0.008890358
1993.154 0.009159685 -0.009159685
1993.158 0.011731347 -0.011731347
1993.162 0.011027203 -0.011027203
1993.165 0.008453808 -0.008453808
1993.169 0.010660608 -0.010660608
1993.173 0.010780868 -0.010780868
1993.177 0.008852151 -0.008852151
1993.181 0.008487617 -0.008487617
1993.185 0.008308323 -0.008308323
1993.188 0.008158502 -0.008158502
1993.192 0.008149006 -0.008149006
1993.196 0.008657997 -0.008657997
1993.200 0.008632974 -0.008632974
1993.204 0.008280867 -0.008280867
1993.208 0.010380825 -0.010380825

1993.212 0.011166066 -0.011166066
1993.215 0.009375611 -0.009375611
1993.219 0.008397572 -0.008397572
1993.223 0.008705411 -0.008705411
1993.227 0.009628497 -0.009628497
1993.231 0.009775549 -0.009775549
1993.235 0.008747294 -0.008747294
1993.238 0.008772274 -0.008772274
1993.242 0.008830501 -0.008830501
1993.246 0.009304610 -0.009304610
1993.250 0.009303638 -0.009303638
1993.254 0.008687378 -0.008687378
1993.258 0.008346255 -0.008346255
1993.262 0.008112850 -0.008112850
1993.265 0.008115129 -0.008115129
1993.269 0.008175749 -0.008175749
1993.273 0.008396286 -0.008396286
1993.277 0.008500768 -0.008500768
1993.281 0.008291630 -0.008291630
1993.285 0.008120072 -0.008120072
1993.288 0.008104062 -0.008104062
1993.292 0.008177421 -0.008177421
1993.296 0.008587705 -0.008587705
1993.300 0.008500186 -0.008500186
1993.304 0.008152180 -0.008152180
1993.308 0.008132110 -0.008132110
1993.312 0.008155034 -0.008155034
1993.315 0.008222764 -0.008222764
1993.319 0.008179507 -0.008179507
1993.323 0.008482128 -0.008482128
1993.327 0.009185625 -0.009185625
1993.331 0.009159919 -0.009159919
1993.335 0.008760724 -0.008760724
1993.338 0.008744916 -0.008744916
1993.342 0.008498441 -0.008498441
1993.346 0.009050706 -0.009050706
1993.350 0.008924062 -0.008924062
1993.354 0.008234978 -0.008234978
1993.358 0.008151861 -0.008151861
1993.362 0.008117798 -0.008117798
1993.365 0.008568451 -0.008568451
1993.369 0.008708070 -0.008708070
1993.373 0.008301826 -0.008301826

1993.377 0.008216998 -0.008216998
1993.381 0.008211724 -0.008211724
1993.385 0.008693524 -0.008693524
1993.388 0.009302466 -0.009302466
1993.392 0.008732922 -0.008732922
1993.396 0.008104115 -0.008104115
1993.400 0.008160405 -0.008160405
1993.404 0.008182817 -0.008182817
1993.408 0.008190341 -0.008190341
1993.412 0.008178484 -0.008178484
1993.415 0.008711279 -0.008711279
1993.419 0.008619191 -0.008619191
1993.423 0.008130084 -0.008130084
1993.427 0.008589359 -0.008589359
1993.431 0.008558605 -0.008558605
1993.435 0.008282276 -0.008282276
1993.438 0.008560652 -0.008560652
1993.442 0.009305870 -0.009305870
1993.446 0.009199128 -0.009199128
1993.450 0.008851719 -0.008851719
1993.454 0.008514865 -0.008514865
1993.458 0.008118113 -0.008118113
1993.462 0.008431954 -0.008431954
1993.465 0.008941952 -0.008941952
1993.469 0.009186767 -0.009186767
1993.473 0.009735467 -0.009735467
1993.477 0.009115861 -0.009115861
1993.481 0.008334485 -0.008334485
1993.485 0.008308734 -0.008308734
1993.488 0.008321398 -0.008321398
1993.492 0.008457559 -0.008457559
1993.496 0.008463812 -0.008463812
1993.500 0.008954375 -0.008954375
1993.504 0.008692289 -0.008692289
1993.508 0.008135697 -0.008135697
1993.512 0.008126477 -0.008126477
1993.515 0.008168810 -0.008168810
1993.519 0.008184459 -0.008184459
1993.523 0.009072428 -0.009072428
1993.527 0.009341679 -0.009341679
1993.531 0.009324550 -0.009324550
1993.535 0.008852719 -0.008852719
1993.538 0.008189009 -0.008189009

1993.542 0.008262251 -0.008262251
1993.546 0.008282683 -0.008282683
1993.550 0.008323641 -0.008323641
1993.554 0.008248716 -0.008248716
1993.558 0.008413885 -0.008413885
1993.562 0.012137360 -0.012137360
1993.565 0.011498816 -0.011498816
1993.569 0.009218672 -0.009218672
1993.573 0.009062799 -0.009062799
1993.577 0.010088099 -0.010088099
1993.581 0.009776181 -0.009776181
1993.585 0.008360188 -0.008360188
1993.588 0.008305329 -0.008305329
1993.592 0.008163749 -0.008163749
1993.596 0.008104532 -0.008104532
1993.600 0.009202181 -0.009202181
1993.604 0.009018638 -0.009018638
1993.608 0.008166731 -0.008166731
1993.612 0.008156381 -0.008156381
1993.615 0.008117733 -0.008117733
1993.619 0.009757731 -0.009757731
1993.623 0.010291235 -0.010291235
1993.627 0.010563173 -0.010563173
1993.631 0.009765773 -0.009765773
1993.635 0.008646148 -0.008646148
1993.638 0.008676045 -0.008676045
1993.642 0.008967428 -0.008967428
1993.646 0.008728411 -0.008728411
1993.650 0.008211127 -0.008211127
1993.654 0.008212101 -0.008212101
1993.658 0.008194278 -0.008194278
1993.662 0.009050971 -0.009050971
1993.665 0.009556727 -0.009556727
1993.669 0.008737560 -0.008737560
1993.673 0.008124092 -0.008124092
1993.677 0.008183466 -0.008183466
1993.681 0.008580835 -0.008580835
1993.685 0.008505168 -0.008505168
1993.688 0.008947678 -0.008947678
1993.692 0.011177820 -0.011177820
1993.696 0.010385088 -0.010385088
1993.700 0.008281470 -0.008281470
1993.704 0.008482977 -0.008482977

1993.708 0.008850338 -0.008850338
1993.712 0.008502940 -0.008502940
1993.715 0.008814309 -0.008814309
1993.719 0.009579754 -0.009579754
1993.723 0.009097081 -0.009097081
1993.727 0.008450642 -0.008450642
1993.731 0.009638579 -0.009638579
1993.735 0.009497427 -0.009497427
1993.738 0.008692032 -0.008692032
1993.742 0.008596052 -0.008596052
1993.746 0.008768712 -0.008768712
1993.750 0.010085346 -0.010085346
1993.754 0.009466051 -0.009466051
1993.758 0.008187839 -0.008187839
1993.762 0.008353270 -0.008353270
1993.765 0.008358302 -0.008358302
1993.769 0.008279123 -0.008279123
1993.773 0.009328844 -0.009328844
1993.777 0.009265629 -0.009265629
1993.781 0.008347700 -0.008347700
1993.785 0.008190971 -0.008190971
1993.788 0.008408499 -0.008408499
1993.792 0.008348158 -0.008348158
1993.796 0.008530368 -0.008530368
1993.800 0.008714788 -0.008714788
1993.804 0.009099761 -0.009099761
1993.808 0.011203409 -0.011203409
1993.812 0.010294893 -0.010294893
1993.815 0.008179028 -0.008179028
1993.819 0.008669508 -0.008669508
1993.823 0.008661613 -0.008661613
1993.827 0.008509861 -0.008509861
1993.831 0.008592511 -0.008592511
1993.835 0.008573181 -0.008573181
1993.838 0.008350431 -0.008350431
1993.842 0.008132204 -0.008132204
1993.846 0.008125308 -0.008125308
1993.850 0.008382403 -0.008382403
1993.854 0.008373271 -0.008373271
1993.858 0.009967398 -0.009967398
1993.862 0.011948677 -0.011948677
1993.865 0.010721224 -0.010721224
1993.869 0.010979753 -0.010979753

1993.873 0.010446297 -0.010446297
1993.877 0.008797887 -0.008797887
1993.881 0.008524367 -0.008524367
1993.885 0.008119895 -0.008119895
1993.888 0.008144734 -0.008144734
1993.892 0.008128997 -0.008128997
1993.896 0.008380306 -0.008380306
1993.900 0.008378999 -0.008378999
1993.904 0.009548827 -0.009548827
1993.908 0.009307172 -0.009307172
1993.912 0.008398335 -0.008398335
1993.915 0.008409076 -0.008409076
1993.919 0.008416783 -0.008416783
1993.923 0.008318104 -0.008318104
1993.927 0.008116236 -0.008116236
1993.931 0.009368032 -0.009368032
1993.935 0.009306783 -0.009306783
1993.938 0.009210641 -0.009210641
1993.942 0.009190187 -0.009190187
1993.946 0.008353768 -0.008353768
1993.950 0.008186900 -0.008186900
1993.954 0.008377831 -0.008377831
1993.958 0.008283631 -0.008283631
1993.962 0.008348582 -0.008348582
1993.965 0.008378953 -0.008378953
1993.969 0.008455263 -0.008455263
1993.973 0.008616626 -0.008616626
1993.977 0.008776481 -0.008776481
1993.981 0.008818128 -0.008818128
1993.985 0.008472605 -0.008472605
1993.988 0.008705068 -0.008705068
1993.992 0.009082724 -0.009082724
1993.996 0.008565399 -0.008565399
1994.000 0.008128783 -0.008128783
1994.004 0.009598551 -0.009598551
1994.008 0.009342575 -0.009342575
1994.012 0.008606482 -0.008606482
1994.015 0.008512017 -0.008512017
1994.019 0.009364073 -0.009364073
1994.023 0.009158072 -0.009158072
1994.027 0.008203193 -0.008203193
1994.031 0.008316452 -0.008316452
1994.035 0.008590683 -0.008590683

1994.038 0.009012603 -0.009012603
1994.042 0.008709300 -0.008709300
1994.046 0.008832299 -0.008832299
1994.050 0.010505447 -0.010505447
1994.054 0.010933047 -0.010933047
1994.058 0.009589442 -0.009589442
1994.062 0.008981363 -0.008981363
1994.065 0.008709982 -0.008709982
1994.069 0.008215288 -0.008215288
1994.073 0.008150674 -0.008150674
1994.077 0.008138672 -0.008138672
1994.081 0.008639215 -0.008639215
1994.085 0.009026880 -0.009026880
1994.088 0.008535473 -0.008535473
1994.092 0.009681150 -0.009681150
1994.096 0.010353417 -0.010353417
1994.100 0.009618215 -0.009618215
1994.104 0.008977999 -0.008977999
1994.108 0.008604272 -0.008604272
1994.112 0.008467109 -0.008467109
1994.115 0.012120776 -0.012120776
1994.119 0.011453666 -0.011453666
1994.123 0.009585299 -0.009585299
1994.127 0.010572370 -0.010572370
1994.131 0.009288436 -0.009288436
1994.135 0.011321150 -0.011321150
1994.138 0.010814930 -0.010814930
1994.142 0.008104062 -0.008104062
1994.146 0.013218427 -0.013218427
1994.150 0.012488987 -0.012488987
1994.154 0.008825688 -0.008825688
1994.158 0.008852610 -0.008852610
1994.162 0.009862651 -0.009862651
1994.165 0.012618668 -0.012618668
1994.169 0.011650690 -0.011650690
1994.173 0.010114853 -0.010114853
1994.177 0.010659406 -0.010659406
1994.181 0.013812743 -0.013812743
1994.185 0.013078763 -0.013078763
1994.188 0.012032356 -0.012032356
1994.192 0.012710647 -0.012710647
1994.196 0.010322371 -0.010322371
1994.200 0.009972630 -0.009972630

1994.204 0.009526172 -0.009526172
1994.208 0.008940262 -0.008940262
1994.212 0.010082887 -0.010082887
1994.215 0.010318066 -0.010318066
1994.219 0.009426012 -0.009426012
1994.223 0.008408330 -0.008408330
1994.227 0.009840011 -0.009840011
1994.231 0.010791780 -0.010791780
1994.235 0.009656912 -0.009656912
1994.238 0.009393782 -0.009393782
1994.242 0.008954454 -0.008954454
1994.246 0.008487442 -0.008487442
1994.250 0.009252132 -0.009252132
1994.254 0.009894463 -0.009894463
1994.258 0.009201877 -0.009201877
1994.262 0.008624447 -0.008624447
1994.265 0.008424746 -0.008424746
1994.269 0.008104062 -0.008104062
1994.273 0.008263342 -0.008263342
1994.277 0.008743889 -0.008743889
1994.281 0.009381061 -0.009381061
1994.285 0.008981608 -0.008981608
1994.288 0.008336233 -0.008336233
1994.292 0.008207281 -0.008207281
1994.296 0.008213952 -0.008213952
1994.300 0.009245901 -0.009245901
1994.304 0.009721843 -0.009721843
1994.308 0.008781659 -0.008781659
1994.312 0.009073693 -0.009073693
1994.315 0.009205047 -0.009205047
1994.319 0.009654258 -0.009654258
1994.323 0.009230692 -0.009230692
1994.327 0.008737365 -0.008737365
1994.331 0.008759557 -0.008759557
1994.335 0.008472671 -0.008472671
1994.338 0.008794055 -0.008794055
1994.342 0.009540048 -0.009540048
1994.346 0.009756804 -0.009756804
1994.350 0.008889340 -0.008889340
1994.354 0.010550506 -0.010550506
1994.358 0.010583308 -0.010583308
1994.362 0.010224448 -0.010224448
1994.365 0.013806572 -0.013806572

1994.369 0.014700456 -0.014700456
1994.373 0.011058001 -0.011058001
1994.377 0.008119368 -0.008119368
1994.381 0.010219497 -0.010219497
1994.385 0.010679393 -0.010679393
1994.388 0.009223505 -0.009223505
1994.392 0.008580916 -0.008580916
1994.396 0.008493170 -0.008493170
1994.400 0.008346414 -0.008346414
1994.404 0.008146318 -0.008146318
1994.408 0.009754607 -0.009754607
1994.412 0.009816570 -0.009816570
1994.415 0.008702251 -0.008702251
1994.419 0.008876891 -0.008876891
1994.423 0.009496253 -0.009496253
1994.427 0.009304104 -0.009304104
1994.431 0.008528270 -0.008528270
1994.435 0.008203383 -0.008203383
1994.438 0.008138692 -0.008138692
1994.442 0.010687380 -0.010687380
1994.446 0.010277445 -0.010277445
1994.450 0.008644431 -0.008644431
1994.454 0.009087971 -0.009087971
1994.458 0.008753671 -0.008753671
1994.462 0.008649834 -0.008649834
1994.465 0.008451305 -0.008451305
1994.469 0.009577649 -0.009577649
1994.473 0.010457786 -0.010457786
1994.477 0.010609811 -0.010609811
1994.481 0.014917947 -0.014917947
1994.485 0.013239722 -0.013239722
1994.488 0.010874683 -0.010874683
1994.492 0.011281492 -0.011281492
1994.496 0.011322494 -0.011322494
1994.500 0.010464753 -0.010464753
1994.504 0.010296686 -0.010296686
1994.508 0.010448603 -0.010448603
1994.512 0.009300799 -0.009300799
1994.515 0.009011351 -0.009011351
1994.519 0.010735339 -0.010735339
1994.523 0.010712269 -0.010712269
1994.527 0.008891495 -0.008891495
1994.531 0.008300325 -0.008300325

1994.535 0.009208844 -0.009208844
1994.538 0.008974720 -0.008974720
1994.542 0.011236280 -0.011236280
1994.546 0.011789753 -0.011789753
1994.550 0.012069140 -0.012069140
1994.554 0.010912121 -0.010912121
1994.558 0.008787691 -0.008787691
1994.562 0.009365951 -0.009365951
1994.565 0.010563847 -0.010563847
1994.569 0.009950410 -0.009950410
1994.573 0.008836501 -0.008836501
1994.577 0.008487370 -0.008487370
1994.581 0.008120176 -0.008120176
1994.585 0.011651515 -0.011651515
1994.588 0.011279433 -0.011279433
1994.592 0.008785628 -0.008785628
1994.596 0.008503885 -0.008503885
1994.600 0.010691446 -0.010691446
1994.604 0.010350141 -0.010350141
1994.608 0.008215563 -0.008215563
1994.612 0.008377309 -0.008377309
1994.615 0.008623199 -0.008623199
1994.619 0.009480740 -0.009480740
1994.623 0.009089090 -0.009089090
1994.627 0.008146527 -0.008146527
1994.631 0.008193535 -0.008193535
1994.635 0.008289624 -0.008289624
1994.638 0.008585328 -0.008585328
1994.642 0.008552825 -0.008552825
1994.646 0.008534820 -0.008534820
1994.650 0.009120204 -0.009120204
1994.654 0.009415655 -0.009415655
1994.658 0.008704138 -0.008704138
1994.662 0.008235972 -0.008235972
1994.665 0.008332656 -0.008332656
1994.669 0.009244642 -0.009244642
1994.673 0.011681737 -0.011681737
1994.677 0.010562357 -0.010562357
1994.681 0.008217035 -0.008217035
1994.685 0.008529665 -0.008529665
1994.688 0.010280014 -0.010280014
1994.692 0.009740074 -0.009740074
1994.696 0.008647696 -0.008647696

1994.700 0.008658146 -0.008658146
1994.704 0.008202130 -0.008202130
1994.708 0.008643836 -0.008643836
1994.712 0.008548583 -0.008548583
1994.715 0.008386818 -0.008386818
1994.719 0.008820579 -0.008820579
1994.723 0.008605779 -0.008605779
1994.727 0.008899649 -0.008899649
1994.731 0.008844451 -0.008844451
1994.735 0.008764640 -0.008764640
1994.738 0.008553181 -0.008553181
1994.742 0.008199239 -0.008199239
1994.746 0.008226612 -0.008226612
1994.750 0.008749117 -0.008749117
1994.754 0.008637636 -0.008637636
1994.758 0.008226826 -0.008226826
1994.762 0.009338653 -0.009338653
1994.765 0.009670931 -0.009670931
1994.769 0.010003759 -0.010003759
1994.773 0.009864864 -0.009864864
1994.777 0.010572213 -0.010572213
1994.781 0.009824448 -0.009824448
1994.785 0.008312788 -0.008312788
1994.788 0.011565177 -0.011565177
1994.792 0.011184602 -0.011184602
1994.796 0.008450685 -0.008450685
1994.800 0.009221555 -0.009221555
1994.804 0.009058809 -0.009058809
1994.808 0.008279561 -0.008279561
1994.812 0.009439034 -0.009439034
1994.815 0.009323771 -0.009323771
1994.819 0.008270101 -0.008270101
1994.823 0.008714896 -0.008714896
1994.827 0.008932244 -0.008932244
1994.831 0.009469934 -0.009469934
1994.835 0.009382287 -0.009382287
1994.838 0.008856396 -0.008856396
1994.842 0.009671280 -0.009671280
1994.846 0.009428599 -0.009428599
1994.850 0.008379894 -0.008379894
1994.854 0.008130782 -0.008130782
1994.858 0.009536959 -0.009536959
1994.862 0.009326735 -0.009326735

1994.865 0.008489367 -0.008489367
1994.869 0.008622052 -0.008622052
1994.873 0.010576601 -0.010576601
1994.877 0.010191370 -0.010191370
1994.881 0.008279581 -0.008279581
1994.885 0.008184333 -0.008184333
1994.888 0.008152009 -0.008152009
1994.892 0.008462825 -0.008462825
1994.896 0.008481398 -0.008481398
1994.900 0.008211278 -0.008211278
1994.904 0.008134962 -0.008134962
1994.908 0.009480047 -0.009480047
1994.912 0.009936484 -0.009936484
1994.915 0.009308367 -0.009308367
1994.919 0.008619365 -0.008619365
1994.923 0.008176942 -0.008176942
1994.927 0.008185243 -0.008185243
1994.931 0.008255853 -0.008255853
1994.935 0.008621938 -0.008621938
1994.938 0.008453478 -0.008453478
1994.942 0.009356365 -0.009356365
1994.946 0.009637336 -0.009637336
1994.950 0.008587701 -0.008587701
1994.954 0.008156889 -0.008156889
1994.958 0.008339596 -0.008339596
1994.962 0.008484101 -0.008484101
1994.965 0.008289460 -0.008289460
1994.969 0.008398218 -0.008398218
1994.973 0.008654543 -0.008654543
1994.977 0.008423687 -0.008423687
1994.981 0.008166866 -0.008166866
1994.985 0.008348817 -0.008348817
1994.988 0.008801710 -0.008801710
1994.992 0.008736782 -0.008736782
1994.996 0.008281871 -0.008281871
1995.000 0.008105421 -0.008105421
1995.004 0.008661774 -0.008661774
1995.008 0.008880697 -0.008880697
1995.012 0.009180328 -0.009180328
1995.015 0.008787879 -0.008787879
1995.019 0.008104062 -0.008104062
1995.023 0.008418171 -0.008418171
1995.027 0.008810557 -0.008810557

1995.031 0.008892489 -0.008892489
1995.035 0.008550462 -0.008550462
1995.038 0.008214601 -0.008214601
1995.042 0.008459347 -0.008459347
1995.046 0.008390963 -0.008390963
1995.050 0.008130017 -0.008130017
1995.054 0.008116183 -0.008116183
1995.058 0.008192314 -0.008192314
1995.062 0.008268582 -0.008268582
1995.065 0.008318541 -0.008318541
1995.069 0.008222936 -0.008222936
1995.073 0.008619191 -0.008619191
1995.077 0.011028759 -0.011028759
1995.081 0.010376988 -0.010376988
1995.085 0.008304165 -0.008304165
1995.088 0.009188413 -0.009188413
1995.092 0.008943131 -0.008943131
1995.096 0.008292655 -0.008292655
1995.100 0.008268211 -0.008268211
1995.104 0.008795874 -0.008795874
1995.108 0.008676961 -0.008676961
1995.112 0.008879070 -0.008879070
1995.115 0.009803718 -0.009803718
1995.119 0.009094534 -0.009094534
1995.123 0.008276079 -0.008276079
1995.127 0.008639613 -0.008639613
1995.131 0.008463873 -0.008463873
1995.135 0.008106626 -0.008106626
1995.138 0.008151760 -0.008151760
1995.142 0.008190079 -0.008190079
1995.146 0.008400640 -0.008400640
1995.150 0.008766725 -0.008766725
1995.154 0.008500546 -0.008500546
1995.158 0.008299112 -0.008299112
1995.162 0.008253252 -0.008253252
1995.165 0.008659398 -0.008659398
1995.169 0.008647658 -0.008647658
1995.173 0.008465120 -0.008465120
1995.177 0.008400590 -0.008400590
1995.181 0.008559651 -0.008559651
1995.185 0.008445544 -0.008445544
1995.188 0.009163201 -0.009163201
1995.192 0.010129880 -0.010129880

1995.196 0.009261317 -0.009261317
1995.200 0.009089157 -0.009089157
1995.204 0.011853968 -0.011853968
1995.208 0.010846860 -0.010846860
1995.212 0.008538754 -0.008538754
1995.215 0.008644098 -0.008644098
1995.219 0.008463048 -0.008463048
1995.223 0.008602794 -0.008602794
1995.227 0.008440495 -0.008440495
1995.231 0.008566595 -0.008566595
1995.235 0.008591669 -0.008591669
1995.238 0.008235618 -0.008235618
1995.242 0.008682021 -0.008682021
1995.246 0.010837750 -0.010837750
1995.250 0.010162366 -0.010162366
1995.254 0.008800090 -0.008800090
1995.258 0.008621957 -0.008621957
1995.262 0.009785113 -0.009785113
1995.265 0.010119976 -0.010119976
1995.269 0.008699214 -0.008699214
1995.273 0.008871755 -0.008871755
1995.277 0.008735197 -0.008735197
1995.281 0.008578584 -0.008578584
1995.285 0.008507490 -0.008507490
1995.288 0.008118872 -0.008118872
1995.292 0.008112150 -0.008112150
1995.296 0.008108308 -0.008108308
1995.300 0.008109983 -0.008109983
1995.304 0.008108516 -0.008108516
1995.308 0.008104062 -0.008104062
1995.312 0.008317057 -0.008317057
1995.315 0.009343125 -0.009343125
1995.319 0.009818864 -0.009818864
1995.323 0.009154706 -0.009154706
1995.327 0.008422591 -0.008422591
1995.331 0.008189082 -0.008189082
1995.335 0.008357007 -0.008357007
1995.338 0.008345230 -0.008345230
1995.342 0.009190961 -0.009190961
1995.346 0.008965069 -0.008965069
1995.350 0.008771326 -0.008771326
1995.354 0.008664096 -0.008664096
1995.358 0.008711856 -0.008711856

1995.362 0.008683943 -0.008683943
1995.365 0.008254921 -0.008254921
1995.369 0.008180214 -0.008180214
1995.373 0.008151976 -0.008151976
1995.377 0.009733655 -0.009733655
1995.381 0.010112127 -0.010112127
1995.385 0.008860333 -0.008860333
1995.388 0.008267881 -0.008267881
1995.392 0.008203967 -0.008203967
1995.396 0.008300636 -0.008300636
1995.400 0.008969234 -0.008969234
1995.404 0.010975600 -0.010975600
1995.408 0.010105413 -0.010105413
1995.412 0.008900044 -0.008900044
1995.415 0.008762210 -0.008762210
1995.419 0.008830251 -0.008830251
1995.423 0.008819121 -0.008819121
1995.427 0.008496692 -0.008496692
1995.431 0.008396451 -0.008396451
1995.435 0.008603383 -0.008603383
1995.438 0.008489306 -0.008489306
1995.442 0.008109320 -0.008109320
1995.446 0.008123876 -0.008123876
1995.450 0.008219441 -0.008219441
1995.454 0.008349552 -0.008349552
1995.458 0.008372964 -0.008372964
1995.462 0.008264720 -0.008264720
1995.465 0.008147426 -0.008147426
1995.469 0.008113791 -0.008113791
1995.473 0.008115231 -0.008115231
1995.477 0.008723303 -0.008723303
1995.481 0.009327174 -0.009327174
1995.485 0.008735452 -0.008735452
1995.488 0.008236870 -0.008236870
1995.492 0.008880274 -0.008880274
1995.496 0.008695091 -0.008695091
1995.500 0.008184493 -0.008184493
1995.504 0.008364758 -0.008364758
1995.508 0.008349583 -0.008349583
1995.512 0.008237028 -0.008237028
1995.515 0.008568333 -0.008568333
1995.519 0.008441673 -0.008441673
1995.523 0.008499678 -0.008499678

1995.527 0.008440776 -0.008440776
1995.531 0.008215347 -0.008215347
1995.535 0.008397242 -0.008397242
1995.538 0.008309094 -0.008309094
1995.542 0.008148021 -0.008148021
1995.546 0.008528525 -0.008528525
1995.550 0.008444552 -0.008444552
1995.554 0.008145577 -0.008145577
1995.558 0.008910467 -0.008910467
1995.562 0.009009298 -0.009009298
1995.565 0.008671346 -0.008671346
1995.569 0.009440462 -0.009440462
1995.573 0.009062653 -0.009062653
1995.577 0.008197837 -0.008197837
1995.581 0.008149371 -0.008149371
1995.585 0.008150981 -0.008150981
1995.588 0.008140598 -0.008140598
1995.592 0.008179702 -0.008179702
1995.596 0.008201941 -0.008201941
1995.600 0.008136845 -0.008136845
1995.604 0.008731154 -0.008731154
1995.608 0.009893847 -0.009893847
1995.612 0.009266096 -0.009266096
1995.615 0.008190502 -0.008190502
1995.619 0.008150110 -0.008150110
1995.623 0.008377201 -0.008377201
1995.627 0.008767797 -0.008767797
1995.631 0.008775033 -0.008775033
1995.635 0.008519998 -0.008519998
1995.638 0.009161156 -0.009161156
1995.642 0.010359174 -0.010359174
1995.646 0.009438257 -0.009438257
1995.650 0.008191470 -0.008191470
1995.654 0.008183153 -0.008183153
1995.658 0.008299687 -0.008299687
1995.662 0.008275009 -0.008275009
1995.665 0.008206895 -0.008206895
1995.669 0.008221183 -0.008221183
1995.673 0.008245169 -0.008245169
1995.677 0.008206522 -0.008206522
1995.681 0.008855740 -0.008855740
1995.685 0.008758059 -0.008758059
1995.688 0.008226219 -0.008226219

1995.692 0.009208440 -0.009208440
1995.696 0.009245624 -0.009245624
1995.700 0.009041719 -0.009041719
1995.704 0.008779346 -0.008779346
1995.708 0.008185170 -0.008185170
1995.712 0.008111675 -0.008111675
1995.715 0.008126230 -0.008126230
1995.719 0.008713467 -0.008713467
1995.723 0.008619874 -0.008619874
1995.727 0.008290459 -0.008290459
1995.731 0.008498612 -0.008498612
1995.735 0.008858004 -0.008858004
1995.738 0.008576715 -0.008576715
1995.742 0.008207511 -0.008207511
1995.746 0.009758119 -0.009758119
1995.750 0.009751818 -0.009751818
1995.754 0.009150183 -0.009150183
1995.758 0.009723521 -0.009723521
1995.762 0.009046551 -0.009046551
1995.765 0.008342697 -0.008342697
1995.769 0.008742890 -0.008742890
1995.773 0.008599756 -0.008599756
1995.777 0.008309305 -0.008309305
1995.781 0.008215175 -0.008215175
1995.785 0.008363590 -0.008363590
1995.788 0.008401318 -0.008401318
1995.792 0.008340832 -0.008340832
1995.796 0.008678567 -0.008678567
1995.800 0.008562770 -0.008562770
1995.804 0.008709300 -0.008709300
1995.808 0.008581552 -0.008581552
1995.812 0.008151731 -0.008151731
1995.815 0.008544668 -0.008544668
1995.819 0.008552085 -0.008552085
1995.823 0.008197832 -0.008197832
1995.827 0.012299655 -0.012299655
1995.831 0.011905596 -0.011905596
1995.835 0.008625325 -0.008625325
1995.838 0.008296529 -0.008296529
1995.842 0.010282641 -0.010282641
1995.846 0.010601685 -0.010601685
1995.850 0.010578015 -0.010578015
1995.854 0.009991065 -0.009991065

1995.858 0.008401020 -0.008401020
1995.862 0.008285847 -0.008285847
1995.865 0.008405735 -0.008405735
1995.869 0.008333786 -0.008333786
1995.873 0.008180975 -0.008180975
1995.877 0.008351899 -0.008351899
1995.881 0.008611731 -0.008611731
1995.885 0.008425504 -0.008425504
1995.888 0.008268472 -0.008268472
1995.892 0.008249434 -0.008249434
1995.896 0.008592126 -0.008592126
1995.900 0.008484078 -0.008484078
1995.904 0.008795917 -0.008795917
1995.908 0.008669463 -0.008669463
1995.912 0.008334406 -0.008334406
1995.915 0.008343564 -0.008343564
1995.919 0.008239179 -0.008239179
1995.923 0.008879786 -0.008879786
1995.927 0.009013212 -0.009013212
1995.931 0.008588632 -0.008588632
1995.935 0.008296703 -0.008296703
1995.938 0.008194159 -0.008194159
1995.942 0.008380440 -0.008380440
1995.946 0.008285450 -0.008285450
1995.950 0.008110822 -0.008110822
1995.954 0.009028731 -0.009028731
1995.958 0.008866430 -0.008866430
1995.962 0.008274395 -0.008274395
1995.965 0.008515592 -0.008515592
1995.969 0.008372640 -0.008372640
1995.973 0.008139786 -0.008139786
1995.977 0.008334093 -0.008334093
1995.981 0.009307234 -0.009307234
1995.985 0.009731891 -0.009731891
1995.988 0.009706820 -0.009706820
1995.992 0.008912087 -0.008912087
1995.996 0.008492046 -0.008492046
1996.000 0.008423549 -0.008423549
1996.004 0.008104062 -0.008104062
1996.008 0.009174747 -0.009174747
1996.012 0.009239190 -0.009239190
1996.015 0.008328996 -0.008328996
1996.019 0.008104062 -0.008104062

1996.023 0.008104062 -0.008104062
1996.027 0.012411251 -0.012411251
1996.031 0.011817312 -0.011817312
1996.035 0.008443979 -0.008443979
1996.038 0.008369550 -0.008369550
1996.042 0.008172157 -0.008172157
1996.046 0.009576551 -0.009576551
1996.050 0.009901325 -0.009901325
1996.054 0.008712891 -0.008712891
1996.058 0.010451597 -0.010451597
1996.062 0.010306608 -0.010306608
1996.065 0.009267079 -0.009267079
1996.069 0.009853835 -0.009853835
1996.073 0.009007338 -0.009007338
1996.077 0.008125672 -0.008125672
1996.081 0.008816326 -0.008816326
1996.085 0.008691832 -0.008691832
1996.088 0.008186670 -0.008186670
1996.092 0.008470126 -0.008470126
1996.096 0.008389200 -0.008389200
1996.100 0.009219752 -0.009219752
1996.104 0.009105028 -0.009105028
1996.108 0.008892790 -0.008892790
1996.112 0.008757097 -0.008757097
1996.115 0.009779848 -0.009779848
1996.119 0.010115990 -0.010115990
1996.123 0.008787077 -0.008787077
1996.127 0.008372108 -0.008372108
1996.131 0.008291383 -0.008291383
1996.135 0.008191697 -0.008191697
1996.138 0.008208666 -0.008208666
1996.142 0.008161788 -0.008161788
1996.146 0.008154436 -0.008154436
1996.150 0.008176187 -0.008176187
1996.154 0.008214656 -0.008214656
1996.158 0.008699466 -0.008699466
1996.162 0.008615868 -0.008615868
1996.165 0.008258138 -0.008258138
1996.169 0.008904674 -0.008904674
1996.173 0.008708717 -0.008708717
1996.177 0.008201447 -0.008201447
1996.181 0.009852152 -0.009852152
1996.185 0.009527986 -0.009527986

1996.188 0.008766320 -0.008766320
1996.192 0.008785414 -0.008785414
1996.196 0.008436202 -0.008436202
1996.200 0.008687723 -0.008687723
1996.204 0.019863047 -0.019863047
1996.208 0.018479864 -0.018479864
1996.212 0.009712832 -0.009712832
1996.215 0.009716787 -0.009716787
1996.219 0.008896621 -0.008896621
1996.223 0.008516532 -0.008516532
1996.227 0.008494654 -0.008494654
1996.231 0.008566920 -0.008566920
1996.235 0.008638375 -0.008638375
1996.238 0.008359532 -0.008359532
1996.242 0.008323551 -0.008323551
1996.246 0.008469882 -0.008469882
1996.250 0.008763681 -0.008763681
1996.254 0.008719442 -0.008719442
1996.258 0.008375625 -0.008375625
1996.262 0.008731526 -0.008731526
1996.265 0.008580899 -0.008580899
1996.269 0.008159939 -0.008159939
1996.273 0.008155968 -0.008155968
1996.277 0.009120438 -0.009120438
1996.281 0.008944257 -0.008944257
1996.285 0.008113909 -0.008113909
1996.288 0.008104062 -0.008104062
1996.292 0.008275781 -0.008275781
1996.296 0.008276108 -0.008276108
1996.300 0.008848364 -0.008848364
1996.304 0.008743388 -0.008743388
1996.308 0.008265991 -0.008265991
1996.312 0.008401132 -0.008401132
1996.315 0.008274331 -0.008274331
1996.319 0.008115198 -0.008115198
1996.323 0.008543195 -0.008543195
1996.327 0.009571071 -0.009571071
1996.331 0.009414212 -0.009414212
1996.335 0.008683949 -0.008683949
1996.338 0.008321750 -0.008321750
1996.342 0.008555273 -0.008555273
1996.346 0.009256246 -0.009256246
1996.350 0.008805520 -0.008805520

1996.354 0.008129748 -0.008129748
1996.358 0.008243437 -0.008243437
1996.362 0.008577841 -0.008577841
1996.365 0.008631622 -0.008631622
1996.369 0.008337196 -0.008337196
1996.373 0.009567450 -0.009567450
1996.377 0.009386377 -0.009386377
1996.381 0.008946909 -0.008946909
1996.385 0.008739088 -0.008739088
1996.388 0.008479280 -0.008479280
1996.392 0.008472267 -0.008472267
1996.396 0.008154665 -0.008154665
1996.400 0.008364311 -0.008364311
1996.404 0.008367593 -0.008367593
1996.408 0.008152452 -0.008152452
1996.412 0.008948088 -0.008948088
1996.415 0.008802449 -0.008802449
1996.419 0.008490410 -0.008490410
1996.423 0.008419627 -0.008419627
1996.427 0.008188514 -0.008188514
1996.431 0.008575123 -0.008575123
1996.435 0.008692958 -0.008692958
1996.438 0.008531438 -0.008531438
1996.442 0.008363043 -0.008363043
1996.446 0.008515809 -0.008515809
1996.450 0.008397803 -0.008397803
1996.454 0.008225986 -0.008225986
1996.458 0.009041733 -0.009041733
1996.462 0.009254591 -0.009254591
1996.465 0.008501059 -0.008501059
1996.469 0.008496121 -0.008496121
1996.473 0.008452370 -0.008452370
1996.477 0.008238050 -0.008238050
1996.481 0.008220401 -0.008220401
1996.485 0.008600273 -0.008600273
1996.488 0.008633332 -0.008633332
1996.492 0.008303416 -0.008303416
1996.496 0.008756836 -0.008756836
1996.500 0.008701771 -0.008701771
1996.504 0.008203507 -0.008203507
1996.508 0.008433995 -0.008433995
1996.512 0.008609329 -0.008609329
1996.515 0.008798680 -0.008798680

1996.519 0.008534163 -0.008534163
1996.523 0.008404032 -0.008404032
1996.527 0.008364027 -0.008364027
1996.531 0.008237608 -0.008237608
1996.535 0.009065623 -0.009065623
1996.538 0.008830441 -0.008830441
1996.542 0.009000834 -0.009000834
1996.546 0.008955473 -0.008955473
1996.550 0.008465698 -0.008465698
1996.554 0.008636769 -0.008636769
1996.558 0.008862682 -0.008862682
1996.562 0.011370429 -0.011370429
1996.565 0.010808231 -0.010808231
1996.569 0.008365891 -0.008365891
1996.573 0.008219039 -0.008219039
1996.577 0.012575035 -0.012575035
1996.581 0.012012075 -0.012012075
1996.585 0.013382857 -0.013382857
1996.588 0.013717016 -0.013717016
1996.592 0.009584657 -0.009584657
1996.596 0.008191717 -0.008191717
1996.600 0.008781614 -0.008781614
1996.604 0.008951846 -0.008951846
1996.608 0.008403913 -0.008403913
1996.612 0.013851946 -0.013851946
1996.615 0.013143514 -0.013143514
1996.619 0.008299142 -0.008299142
1996.623 0.008778393 -0.008778393
1996.627 0.008696493 -0.008696493
1996.631 0.008208449 -0.008208449
1996.635 0.008397757 -0.008397757
1996.638 0.008334616 -0.008334616
1996.642 0.008380852 -0.008380852
1996.646 0.008323038 -0.008323038
1996.650 0.008459327 -0.008459327
1996.654 0.008410648 -0.008410648
1996.658 0.008775462 -0.008775462
1996.662 0.008759291 -0.008759291
1996.665 0.008687424 -0.008687424
1996.669 0.008592323 -0.008592323
1996.673 0.008244600 -0.008244600
1996.677 0.008194958 -0.008194958
1996.681 0.008637473 -0.008637473

1996.685 0.008577842 -0.008577842
1996.688 0.008905717 -0.008905717
1996.692 0.008769917 -0.008769917
1996.696 0.008696034 -0.008696034
1996.700 0.008627014 -0.008627014
1996.704 0.008187910 -0.008187910
1996.708 0.008198936 -0.008198936
1996.712 0.009371081 -0.009371081
1996.715 0.009165456 -0.009165456
1996.719 0.008287863 -0.008287863
1996.723 0.008257791 -0.008257791
1996.727 0.008636580 -0.008636580
1996.731 0.008536601 -0.008536601
1996.735 0.008725596 -0.008725596
1996.738 0.008604572 -0.008604572
1996.742 0.008117703 -0.008117703
1996.746 0.008115255 -0.008115255
1996.750 0.010003002 -0.010003002
1996.754 0.010019502 -0.010019502
1996.758 0.009159100 -0.009159100
1996.762 0.008733789 -0.008733789
1996.765 0.008110596 -0.008110596
1996.769 0.008986375 -0.008986375
1996.773 0.008832224 -0.008832224
1996.777 0.008395529 -0.008395529
1996.781 0.008415137 -0.008415137
1996.785 0.008164577 -0.008164577
1996.788 0.009044690 -0.009044690
1996.792 0.008893146 -0.008893146
1996.796 0.008170399 -0.008170399
1996.800 0.008213515 -0.008213515
1996.804 0.008189997 -0.008189997
1996.808 0.008172612 -0.008172612
1996.812 0.008176768 -0.008176768
1996.815 0.008686850 -0.008686850
1996.819 0.008564556 -0.008564556
1996.823 0.008165091 -0.008165091
1996.827 0.008276905 -0.008276905
1996.831 0.008300556 -0.008300556
1996.835 0.008755050 -0.008755050
1996.838 0.008644293 -0.008644293
1996.842 0.008172677 -0.008172677
1996.846 0.008129397 -0.008129397

1996.850 0.008665956 -0.008665956
1996.854 0.008603011 -0.008603011
1996.858 0.008144566 -0.008144566
1996.862 0.008173761 -0.008173761
1996.865 0.008257115 -0.008257115
1996.869 0.008870385 -0.008870385
1996.873 0.009152013 -0.009152013
1996.877 0.008531059 -0.008531059
1996.881 0.008111131 -0.008111131
1996.885 0.008237783 -0.008237783
1996.888 0.008216503 -0.008216503
1996.892 0.008115509 -0.008115509
1996.896 0.008611721 -0.008611721
1996.900 0.009356134 -0.009356134
1996.904 0.008875851 -0.008875851
1996.908 0.008154079 -0.008154079
1996.912 0.008158191 -0.008158191
1996.915 0.008392811 -0.008392811
1996.919 0.008546449 -0.008546449
1996.923 0.008343434 -0.008343434
1996.927 0.008235273 -0.008235273
1996.931 0.008744976 -0.008744976
1996.935 0.009309583 -0.009309583
1996.938 0.008912267 -0.008912267
1996.942 0.008666519 -0.008666519
1996.946 0.009664675 -0.009664675
1996.950 0.009238214 -0.009238214
1996.954 0.008192430 -0.008192430
1996.958 0.010918351 -0.010918351
1996.962 0.011072969 -0.011072969
1996.965 0.008804724 -0.008804724
1996.969 0.008564655 -0.008564655
1996.973 0.008441920 -0.008441920
1996.977 0.008336807 -0.008336807
1996.981 0.008755590 -0.008755590
1996.985 0.008674505 -0.008674505
1996.988 0.008550582 -0.008550582
1996.992 0.008371969 -0.008371969
1996.996 0.008170448 -0.008170448
1997.000 0.008254829 -0.008254829
1997.004 0.008214524 -0.008214524
1997.008 0.008118409 -0.008118409
1997.012 0.008104062 -0.008104062

1997.015 0.009452875 -0.009452875
1997.019 0.009532520 -0.009532520
1997.023 0.008409198 -0.008409198
1997.027 0.008119892 -0.008119892
1997.031 0.008104062 -0.008104062
1997.035 0.008106355 -0.008106355
1997.038 0.008243544 -0.008243544
1997.042 0.008217603 -0.008217603
1997.046 0.008353180 -0.008353180
1997.050 0.008534149 -0.008534149
1997.054 0.008470317 -0.008470317
1997.058 0.008642215 -0.008642215
1997.062 0.008463085 -0.008463085
1997.065 0.008139596 -0.008139596
1997.069 0.008484829 -0.008484829
1997.073 0.008491301 -0.008491301
1997.077 0.008224781 -0.008224781
1997.081 0.008240012 -0.008240012
1997.085 0.011587210 -0.011587210
1997.088 0.011698267 -0.011698267
1997.092 0.009398489 -0.009398489
1997.096 0.008578338 -0.008578338
1997.100 0.009414703 -0.009414703
1997.104 0.009257129 -0.009257129
1997.108 0.008844915 -0.008844915
1997.112 0.010206039 -0.010206039
1997.115 0.009500910 -0.009500910
1997.119 0.008226227 -0.008226227
1997.123 0.008206390 -0.008206390
1997.127 0.008141049 -0.008141049
1997.131 0.009050326 -0.009050326
1997.135 0.009147953 -0.009147953
1997.138 0.008574881 -0.008574881
1997.142 0.008798874 -0.008798874
1997.146 0.008548023 -0.008548023
1997.150 0.008435199 -0.008435199
1997.154 0.009295020 -0.009295020
1997.158 0.009105195 -0.009105195
1997.162 0.008746630 -0.008746630
1997.165 0.008500421 -0.008500421
1997.169 0.008140216 -0.008140216
1997.173 0.009425634 -0.009425634
1997.177 0.009755992 -0.009755992

1997.181 0.009078431 -0.009078431
1997.185 0.008632911 -0.008632911
1997.188 0.008618404 -0.008618404
1997.192 0.008722377 -0.008722377
1997.196 0.009403004 -0.009403004
1997.200 0.009387364 -0.009387364
1997.204 0.009623788 -0.009623788
1997.208 0.009484367 -0.009484367
1997.212 0.009157426 -0.009157426
1997.215 0.008730939 -0.008730939
1997.219 0.008120467 -0.008120467
1997.223 0.009611078 -0.009611078
1997.227 0.009582226 -0.009582226
1997.231 0.010417032 -0.010417032
1997.235 0.010319598 -0.010319598
1997.238 0.008610406 -0.008610406
1997.242 0.011050534 -0.011050534
1997.246 0.011241358 -0.011241358
1997.250 0.008861222 -0.008861222
1997.254 0.009479494 -0.009479494
1997.258 0.010448930 -0.010448930
1997.262 0.009712163 -0.009712163
1997.265 0.008552773 -0.008552773
1997.269 0.008104062 -0.008104062
1997.273 0.014848695 -0.014848695
1997.277 0.013921908 -0.013921908
1997.281 0.008401936 -0.008401936
1997.285 0.008327803 -0.008327803
1997.288 0.012352568 -0.012352568
1997.292 0.011701634 -0.011701634
1997.296 0.009093016 -0.009093016
1997.300 0.008940448 -0.008940448
1997.304 0.008312908 -0.008312908
1997.308 0.008378110 -0.008378110
1997.312 0.009345989 -0.009345989
1997.315 0.009168136 -0.009168136
1997.319 0.008775774 -0.008775774
1997.323 0.008963325 -0.008963325
1997.327 0.009001194 -0.009001194
1997.331 0.008639940 -0.008639940
1997.335 0.008435901 -0.008435901
1997.338 0.009312564 -0.009312564
1997.342 0.010139942 -0.010139942

1997.346 0.009335258 -0.009335258
1997.350 0.009446179 -0.009446179
1997.354 0.009656644 -0.009656644
1997.358 0.008592825 -0.008592825
1997.362 0.009084984 -0.009084984
1997.365 0.010485315 -0.010485315
1997.369 0.009975190 -0.009975190
1997.373 0.008770946 -0.008770946
1997.377 0.008303403 -0.008303403
1997.381 0.008145040 -0.008145040
1997.385 0.008581027 -0.008581027
1997.388 0.008979841 -0.008979841
1997.392 0.009249321 -0.009249321
1997.396 0.008733702 -0.008733702
1997.400 0.008193352 -0.008193352
1997.404 0.008166222 -0.008166222
1997.408 0.009742170 -0.009742170
1997.412 0.011707175 -0.011707175
1997.415 0.010227018 -0.010227018
1997.419 0.008111628 -0.008111628
1997.423 0.008182850 -0.008182850
1997.427 0.008177851 -0.008177851
1997.431 0.009042310 -0.009042310
1997.435 0.008873381 -0.008873381
1997.438 0.010349714 -0.010349714
1997.442 0.012606519 -0.012606519
1997.446 0.011413420 -0.011413420
1997.450 0.009136999 -0.009136999
1997.454 0.008384620 -0.008384620
1997.458 0.009406690 -0.009406690
1997.462 0.009736190 -0.009736190
1997.465 0.008651581 -0.008651581
1997.469 0.008883618 -0.008883618
1997.473 0.009518886 -0.009518886
1997.477 0.008900824 -0.008900824
1997.481 0.008339035 -0.008339035
1997.485 0.008292172 -0.008292172
1997.488 0.009081373 -0.009081373
1997.492 0.011287016 -0.011287016
1997.496 0.010816665 -0.010816665
1997.500 0.008821601 -0.008821601
1997.504 0.008281775 -0.008281775
1997.508 0.009850152 -0.009850152

1997.512 0.009552282 -0.009552282
1997.515 0.008343092 -0.008343092
1997.519 0.009731445 -0.009731445
1997.523 0.009580852 -0.009580852
1997.527 0.008436867 -0.008436867
1997.531 0.011607427 -0.011607427
1997.535 0.011312818 -0.011312818
1997.538 0.010448963 -0.010448963
1997.542 0.010613129 -0.010613129
1997.546 0.009121180 -0.009121180
1997.550 0.010036669 -0.010036669
1997.554 0.010878596 -0.010878596
1997.558 0.009767249 -0.009767249
1997.562 0.008479220 -0.008479220
1997.565 0.009569982 -0.009569982
1997.569 0.010006241 -0.010006241
1997.573 0.011976140 -0.011976140
1997.577 0.013258764 -0.013258764
1997.581 0.011140654 -0.011140654
1997.585 0.014915457 -0.014915457
1997.588 0.013679807 -0.013679807
1997.592 0.008187497 -0.008187497
1997.596 0.008490164 -0.008490164
1997.600 0.008884775 -0.008884775
1997.604 0.010055948 -0.010055948
1997.608 0.009645362 -0.009645362
1997.612 0.008287659 -0.008287659
1997.615 0.011309225 -0.011309225
1997.619 0.010810809 -0.010810809
1997.623 0.008624252 -0.008624252
1997.627 0.010897068 -0.010897068
1997.631 0.010761828 -0.010761828
1997.635 0.009039873 -0.009039873
1997.638 0.008464764 -0.008464764
1997.642 0.011396643 -0.011396643
1997.646 0.012691119 -0.012691119
1997.650 0.011372882 -0.011372882
1997.654 0.011254378 -0.011254378
1997.658 0.015133044 -0.015133044
1997.662 0.014966827 -0.014966827
1997.665 0.010414282 -0.010414282
1997.669 0.014975233 -0.014975233
1997.673 0.013949820 -0.013949820

1997.677 0.010964967 -0.010964967
1997.681 0.010959457 -0.010959457
1997.685 0.015656091 -0.015656091
1997.688 0.014367498 -0.014367498
1997.692 0.008942730 -0.008942730
1997.696 0.014785200 -0.014785200
1997.700 0.013630653 -0.013630653
1997.704 0.008316166 -0.008316166
1997.708 0.008868294 -0.008868294
1997.712 0.008921537 -0.008921537
1997.715 0.009175419 -0.009175419
1997.719 0.010664676 -0.010664676
1997.723 0.011005336 -0.011005336
1997.727 0.009357355 -0.009357355
1997.731 0.008547373 -0.008547373
1997.735 0.010661149 -0.010661149
1997.738 0.013290070 -0.013290070
1997.742 0.012424774 -0.012424774
1997.746 0.009394788 -0.009394788
1997.750 0.010148295 -0.010148295
1997.754 0.009897269 -0.009897269
1997.758 0.008233884 -0.008233884
1997.762 0.009106324 -0.009106324
1997.765 0.009472122 -0.009472122
1997.769 0.008733599 -0.008733599
1997.773 0.008309592 -0.008309592
1997.777 0.009651329 -0.009651329
1997.781 0.010325564 -0.010325564
1997.785 0.011068456 -0.011068456
1997.788 0.010151804 -0.010151804
1997.792 0.008855831 -0.008855831
1997.796 0.008714852 -0.008714852
1997.800 0.009971293 -0.009971293
1997.804 0.009802646 -0.009802646
1997.808 0.010183600 -0.010183600
1997.812 0.010066157 -0.010066157
1997.815 0.008561179 -0.008561179
1997.819 0.008204510 -0.008204510
1997.823 0.008825090 -0.008825090
1997.827 0.008835108 -0.008835108
1997.831 0.009807119 -0.009807119
1997.835 0.010133299 -0.010133299
1997.838 0.012977350 -0.012977350

1997.842 0.012105364 -0.012105364
1997.846 0.013099921 -0.013099921
1997.850 0.020911474 -0.020911474
1997.854 0.021959810 -0.021959810
1997.858 0.016283689 -0.016283689
1997.862 0.012234323 -0.012234323
1997.865 0.012479788 -0.012479788
1997.869 0.010941939 -0.010941939
1997.873 0.009443686 -0.009443686
1997.877 0.009370747 -0.009370747
1997.881 0.011556086 -0.011556086
1997.885 0.010853719 -0.010853719
1997.888 0.008357464 -0.008357464
1997.892 0.008872180 -0.008872180
1997.896 0.008704179 -0.008704179
1997.900 0.008255745 -0.008255745
1997.904 0.011740835 -0.011740835
1997.908 0.011144743 -0.011144743
1997.912 0.008123749 -0.008123749
1997.915 0.009563004 -0.009563004
1997.919 0.010470575 -0.010470575
1997.923 0.010497326 -0.010497326
1997.927 0.009411526 -0.009411526
1997.931 0.009379326 -0.009379326
1997.935 0.009348780 -0.009348780
1997.938 0.008336388 -0.008336388
1997.942 0.010182169 -0.010182169
1997.946 0.010216864 -0.010216864
1997.950 0.008479664 -0.008479664
1997.954 0.008591237 -0.008591237
1997.958 0.008833185 -0.008833185
1997.962 0.009847185 -0.009847185
1997.965 0.009367001 -0.009367001
1997.969 0.008575920 -0.008575920
1997.973 0.008736930 -0.008736930
1997.977 0.008323332 -0.008323332
1997.981 0.008333541 -0.008333541
1997.985 0.010447447 -0.010447447
1997.988 0.010092865 -0.010092865
1997.992 0.008266603 -0.008266603
1997.996 0.011037894 -0.011037894
1998.000 0.011153322 -0.011153322
1998.004 0.008729998 -0.008729998

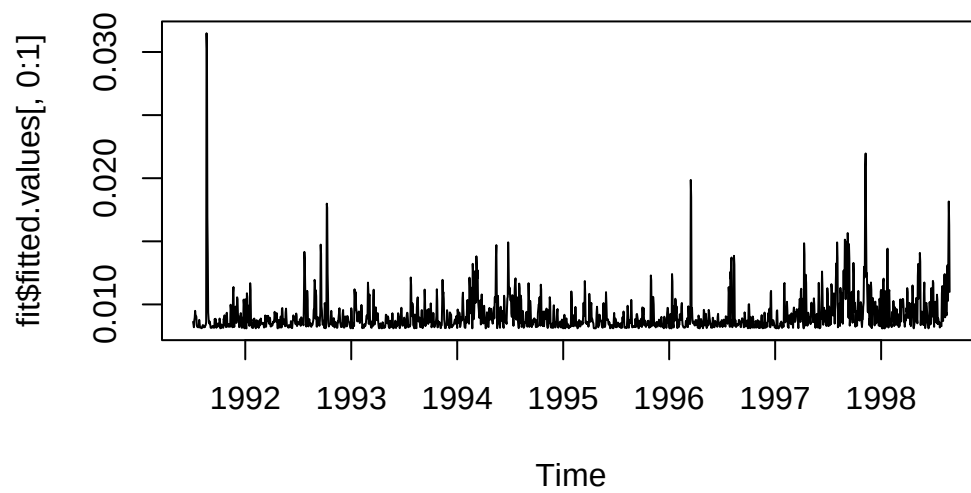
1998.008 0.008108780 -0.008108780
1998.012 0.008104787 -0.008104787
1998.015 0.008104062 -0.008104062
1998.019 0.011742634 -0.011742634
1998.023 0.012043618 -0.012043618
1998.027 0.009066036 -0.009066036
1998.031 0.008104806 -0.008104806
1998.035 0.008104062 -0.008104062
1998.038 0.011052247 -0.011052247
1998.042 0.010652765 -0.010652765
1998.046 0.008217540 -0.008217540
1998.050 0.008819751 -0.008819751
1998.054 0.009772664 -0.009772664
1998.058 0.014375366 -0.014375366
1998.062 0.014418822 -0.014418822
1998.065 0.009999945 -0.009999945
1998.069 0.008170582 -0.008170582
1998.073 0.010914551 -0.010914551
1998.077 0.011137674 -0.011137674
1998.081 0.009386439 -0.009386439
1998.085 0.008627303 -0.008627303
1998.088 0.008353978 -0.008353978
1998.092 0.008537933 -0.008537933
1998.096 0.008377058 -0.008377058
1998.100 0.008201331 -0.008201331
1998.104 0.009411556 -0.009411556
1998.108 0.009249827 -0.009249827
1998.112 0.008669709 -0.008669709
1998.115 0.010272206 -0.010272206
1998.119 0.009792821 -0.009792821
1998.123 0.008284676 -0.008284676
1998.127 0.008802266 -0.008802266
1998.131 0.009772739 -0.009772739
1998.135 0.009205073 -0.009205073
1998.138 0.008396372 -0.008396372
1998.142 0.009048822 -0.009048822
1998.146 0.009630320 -0.009630320
1998.150 0.009148213 -0.009148213
1998.154 0.008364725 -0.008364725
1998.158 0.009244115 -0.009244115
1998.162 0.009133388 -0.009133388
1998.165 0.008211898 -0.008211898
1998.169 0.008311122 -0.008311122

1998.173 0.008255823 -0.008255823
1998.177 0.008380587 -0.008380587
1998.181 0.010396545 -0.010396545
1998.185 0.010236198 -0.010236198
1998.188 0.008647614 -0.008647614
1998.192 0.010215672 -0.010215672
1998.196 0.009788478 -0.009788478
1998.200 0.010398528 -0.010398528
1998.204 0.010370837 -0.010370837
1998.208 0.010423358 -0.010423358
1998.212 0.009809918 -0.009809918
1998.215 0.008638300 -0.008638300
1998.219 0.008667191 -0.008667191
1998.223 0.008229562 -0.008229562
1998.227 0.008643560 -0.008643560
1998.231 0.009161865 -0.009161865
1998.235 0.008729556 -0.008729556
1998.238 0.009716377 -0.009716377
1998.242 0.009410957 -0.009410957
1998.246 0.011295185 -0.011295185
1998.250 0.010966420 -0.010966420
1998.254 0.008920639 -0.008920639
1998.258 0.009180799 -0.009180799
1998.262 0.009014577 -0.009014577
1998.265 0.010123122 -0.010123122
1998.269 0.009536041 -0.009536041
1998.273 0.008443164 -0.008443164
1998.277 0.008500848 -0.008500848
1998.281 0.008280777 -0.008280777
1998.285 0.009138159 -0.009138159
1998.288 0.010095444 -0.010095444
1998.292 0.009955183 -0.009955183
1998.296 0.011504436 -0.011504436
1998.300 0.010640122 -0.010640122
1998.304 0.008251214 -0.008251214
1998.308 0.008104062 -0.008104062
1998.312 0.008309575 -0.008309575
1998.315 0.008562413 -0.008562413
1998.319 0.010030737 -0.010030737
1998.323 0.009803643 -0.009803643
1998.327 0.008651866 -0.008651866
1998.331 0.010509939 -0.010509939
1998.335 0.010384665 -0.010384665

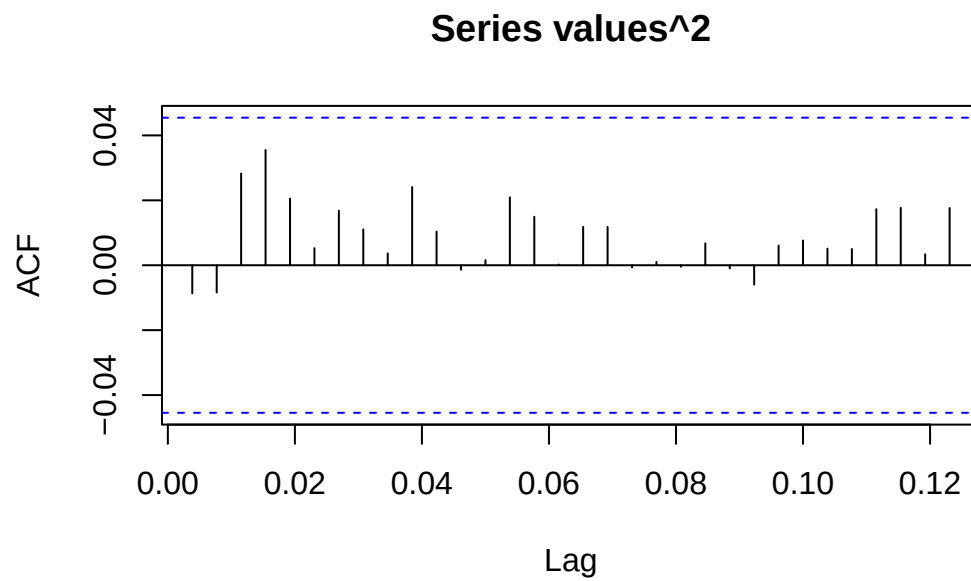
1998.338 0.008806126 -0.008806126
1998.342 0.008498968 -0.008498968
1998.346 0.012245955 -0.012245955
1998.350 0.013207084 -0.013207084
1998.354 0.010452683 -0.010452683
1998.358 0.011651183 -0.011651183
1998.362 0.010804151 -0.010804151
1998.365 0.014082976 -0.014082976
1998.369 0.013380207 -0.013380207
1998.373 0.008357057 -0.008357057
1998.377 0.008879299 -0.008879299
1998.381 0.009062804 -0.009062804
1998.385 0.009431202 -0.009431202
1998.388 0.009294165 -0.009294165
1998.392 0.008647615 -0.008647615
1998.396 0.008469052 -0.008469052
1998.400 0.008358576 -0.008358576
1998.404 0.010920043 -0.010920043
1998.408 0.011738467 -0.011738467
1998.412 0.009507343 -0.009507343
1998.415 0.008122801 -0.008122801
1998.419 0.008412413 -0.008412413
1998.423 0.009975896 -0.009975896
1998.427 0.010118848 -0.010118848
1998.431 0.009864274 -0.009864274
1998.435 0.009226691 -0.009226691
1998.438 0.008546916 -0.008546916
1998.442 0.008396064 -0.008396064
1998.446 0.008104332 -0.008104332
1998.450 0.008152701 -0.008152701
1998.454 0.009052270 -0.009052270
1998.458 0.010195159 -0.010195159
1998.462 0.009528358 -0.009528358
1998.465 0.008408250 -0.008408250
1998.469 0.008722646 -0.008722646
1998.473 0.011287892 -0.011287892
1998.477 0.011208658 -0.011208658
1998.481 0.009579195 -0.009579195
1998.485 0.009043701 -0.009043701
1998.488 0.011865519 -0.011865519
1998.492 0.011695373 -0.011695373
1998.496 0.008856586 -0.008856586
1998.500 0.008220086 -0.008220086

```
1998.504 0.009744095 -0.009744095
1998.508 0.009684299 -0.009684299
1998.512 0.010212897 -0.010212897
1998.515 0.009779282 -0.009779282
1998.519 0.008690317 -0.008690317
1998.523 0.008546519 -0.008546519
1998.527 0.010774308 -0.010774308
1998.531 0.010354426 -0.010354426
1998.535 0.008436940 -0.008436940
1998.538 0.008886338 -0.008886338
1998.542 0.008546259 -0.008546259
1998.546 0.008633621 -0.008633621
1998.550 0.008594077 -0.008594077
1998.554 0.008171252 -0.008171252
1998.558 0.008422268 -0.008422268
1998.562 0.008527121 -0.008527121
1998.565 0.008310435 -0.008310435
1998.569 0.008273770 -0.008273770
1998.573 0.010867866 -0.010867866
1998.577 0.011455012 -0.011455012
1998.581 0.009245049 -0.009245049
1998.585 0.008680526 -0.008680526
1998.588 0.009857526 -0.009857526
1998.592 0.009284694 -0.009284694
1998.596 0.012329269 -0.012329269
1998.600 0.012401266 -0.012401266
1998.604 0.008978960 -0.008978960
1998.608 0.009307896 -0.009307896
1998.612 0.009219007 -0.009219007
1998.615 0.009335406 -0.009335406
1998.619 0.009147868 -0.009147868
1998.623 0.013082903 -0.013082903
1998.627 0.013066590 -0.013066590
1998.631 0.010284863 -0.010284863
1998.635 0.014128526 -0.014128526
1998.638 0.018156125 -0.018156125
1998.642 0.016252704 -0.016252704
1998.646 0.010992686 -0.010992686
```

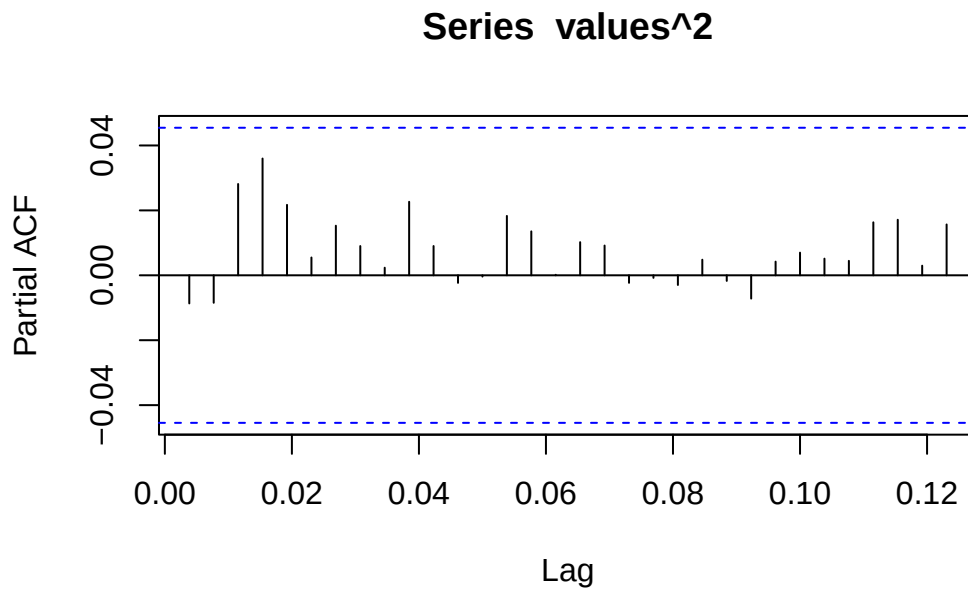
```
plot(fit$fitted.values[,0:1])
```



```
values<-fit$residuals  
acf(values^2,na.action=na.pass)
```



```
pacf(values^2,na.action=na.pass)
```



Fundamentals of Forecasting

Generally speaking, we are looking for the most likely next k values based on the first n observations.

Sources of Uncertainty in Forecasting

Principal sources of uncertainty:

1. Is the model of the past data also appropriate for the future (e.g. change of behavior, policy)?
2. Did you choose the order of the ARMA(p,q) correctly?
3. Are the fitting parameters in the model (sufficiently) accurate?
4. How can you assess the contribution/variability of the innovation terms E_t ?

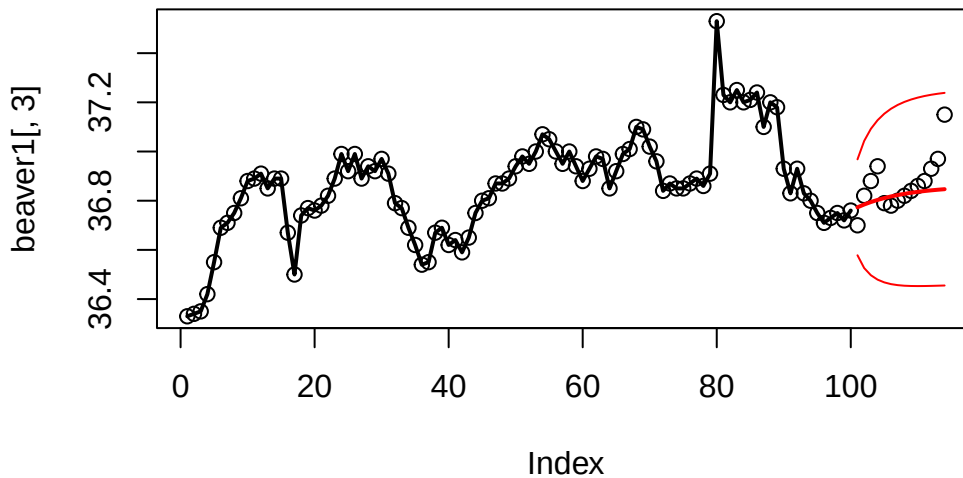
Forecasting different Models

$AR(1)$ processes

For an $AR(1)$ model the forecasted value is just the last value times the fitting coefficient α_1 .

Note: Only the previous value is required.

```
#Extract training set
btrain<-window(beaver1[,3],1,100)
#Fit AR(1) model with Burg's algorithm
fit<-ar.burg(btrain,order=1)
#Perform forecast of 14 steps
forecast <- predict(fit, n.ahead=14)
#Plot forecast
plot(beaver1[,3], lty=3)
lines(btrain, lwd=2)
lines(forecast$pred, lwd=2, col="red")
lines(forecast$pred+forecast$se*1.96, col="red")
lines(forecast$pred-forecast$se*1.96, col="red")
```



$AR(p)$ models

For an $AR(p)$ model the p -step forecast is k steps ahead, the formula is calculated with the previous predictions. The prognosis intervals only consider the uncertainty of the (Gaussian) innovation. Other sources are neglected.

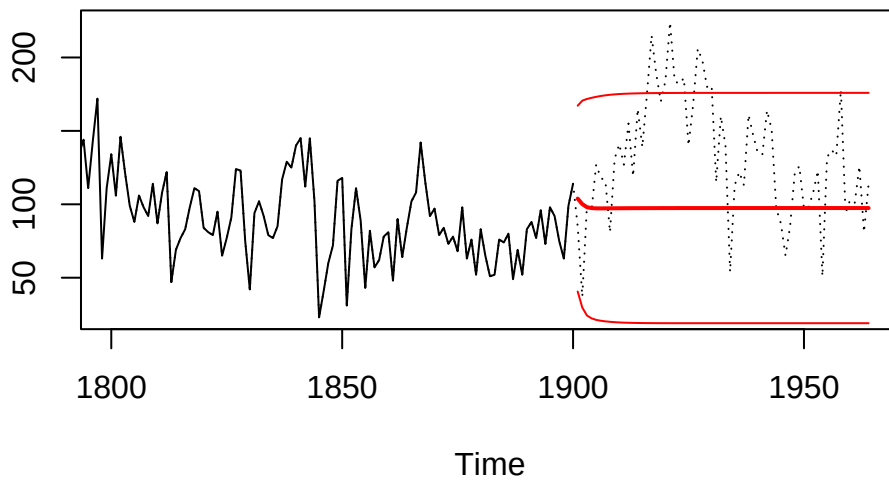
Forecastin *ARIMA* Models

Mathematically derivation of prediction more involved. However, everything is implemented in the predict function.

```
#Read data
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/foehre.dat"
douglasfir <- ts(scan(t.url, skip=1), start=1107)
train <- window(douglasfir, start=1107, end=1900)

#Generate ARMA(4,1) model
fit <- arima(train, order=c(4,0,1))

# Predict 64 steps
fc <- predict(fit, n.ahead=64)
# Plot result
plot(window(douglasfir, 1800, 1964), lty=3, ylab="")
lines(train, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")
```



Forecast for time series with trend and seasonality

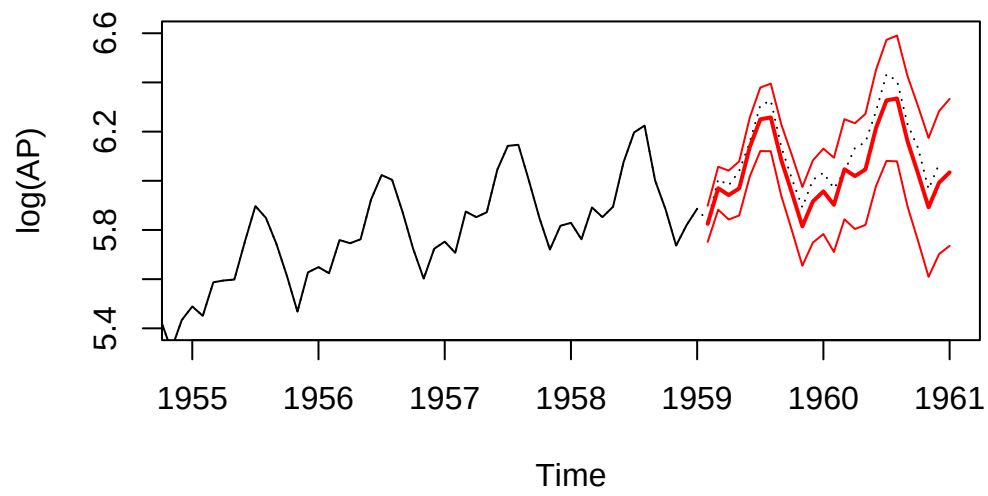
If data are fit by an *SARIMA* model, R takes care of all aspects of the forecast.

```
#Load data
lap<-log(AirPassengers)
ltrain<-window(lap,1949,1959)

#Perform fit
fit<-arima(ltrain,order=c(0,1,1),seasonal=c(0,1,1))

#Perform forecast
fc <- predict(fit, n.ahead=24)

#Plot everything
plot(lap, lty=3,
     xlim=c(1955,1961),ylim=c(5.4,6.6),ylab="log(AP)")
lines(ltrain, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")
```



Features

- The predicted values capture the trend and seasonality.
- The prediction intervals will typically increase over time.
- The prediction quality of the trend is not ascertained and may be poor.